

# BS-CH101

## Chemistry-I

---

### Combined Module Notes

Modules 1 – 7

Semester I • ECE • 4 Credits

CGEC | B.Tech ECE (2023–27)

#### Disclaimer / Study Guide Notice:

These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

**Biraj Sarkar**

IN ASSOCIATION WITH

MAKAUT Wingman • MAKAUT Future Minds

**Cooch Behar Government Engineering College**

*Exam-Ready Notes*

# Contents

---

|                                                                      |           |
|----------------------------------------------------------------------|-----------|
| <b>Module 1: Atomic and Molecular Structure</b>                      | <b>6</b>  |
| <b>1 Atomic and Molecular Structure</b>                              | <b>6</b>  |
| 1.1 Schrodinger Wave Equation                                        | 6         |
| 1.2 Particle in a One-Dimensional Box                                | 6         |
| 1.3 Molecular Orbitals of Diatomic Molecules                         | 7         |
| 1.4 Pi-Molecular Orbitals and Aromaticity                            | 7         |
| 1.5 Pi-Molecular Orbitals of 1,3-Butadiene                           | 8         |
| 1.6 Crystal Field Theory                                             | 8         |
| 1.7 Band Structure and Doping                                        | 9         |
| <b>2 Exam-Oriented Derivations and Applications</b>                  | <b>9</b>  |
| 2.1 Derivation of Particle in a Box Energy                           | 9         |
| 2.2 MO Filling Order and Magnetic Behaviour                          | 10        |
| 2.3 Crystal Field Splitting: High Spin and Low Spin                  | 10        |
| 2.4 n-Type and p-Type Semiconductors                                 | 11        |
| 2.5 Important Long-Answer: Aromaticity of Benzene                    | 11        |
| 2.6 Important Long-Answer: Crystal Field Theory and Magnetism        | 12        |
| 2.7 Band Theory: Exam Numericals and Concepts                        | 13        |
| <b>Module 2: Spectroscopic Techniques and Applications</b>           | <b>16</b> |
| <b>1 Spectroscopic Techniques and Applications</b>                   | <b>16</b> |
| 1.1 Basic Principle and Selection Rules                              | 16        |
| 1.2 Electronic Spectroscopy                                          | 16        |
| 1.3 Fluorescence and Medical Applications                            | 17        |
| 1.4 Vibrational and Rotational Spectroscopy                          | 17        |
| 1.5 NMR, MRI, Surface Characterisation, Diffraction                  | 17        |
| <b>2 Detailed Spectroscopy Notes for Exam Answers</b>                | <b>18</b> |
| 2.1 Electromagnetic Spectrum and Type of Transition                  | 18        |
| 2.2 Beer-Lambert Law in Electronic Spectroscopy                      | 18        |
| 2.3 IR Spectroscopy: Functional Group Region                         | 19        |
| 2.4 NMR Chemical Shift and Splitting                                 | 19        |
| 2.5 Bragg Law Derivation Idea                                        | 19        |
| 2.6 Rotational Spectroscopy of a Rigid Diatomic Molecule             | 20        |
| 2.7 Vibrational Spectroscopy and Hooke's Law Model                   | 20        |
| 2.8 Electronic Spectra: Bathochromic and Hypsochromic Shifts         | 21        |
| 2.9 Fluorescence Versus Phosphorescence                              | 21        |
| 2.10 Short Notes on Surface Characterisation                         | 22        |
| <b>Module 3: Intermolecular Forces and Potential Energy Surfaces</b> | <b>25</b> |
| <b>1 Intermolecular Forces and Potential Energy Surfaces</b>         | <b>25</b> |

|                                                      |                                                                              |           |
|------------------------------------------------------|------------------------------------------------------------------------------|-----------|
| 1.1                                                  | Ionic, Dipole and Van der Waals Interactions . . . . .                       | 25        |
| 1.2                                                  | Potential Energy Curve . . . . .                                             | 25        |
| 1.3                                                  | Equations of State for Real Gases . . . . .                                  | 26        |
| 1.4                                                  | Critical Phenomena . . . . .                                                 | 26        |
| <b>2</b>                                             | <b>Detailed Notes on Forces, Real Gases and Critical Behaviour . . . . .</b> | <b>26</b> |
| 2.1                                                  | Origin and Strength of Intermolecular Forces . . . . .                       | 27        |
| 2.2                                                  | Potential Energy and Force Relation . . . . .                                | 27        |
| 2.3                                                  | Lennard-Jones Potential . . . . .                                            | 27        |
| 2.4                                                  | Compressibility Factor and Deviation from Ideality . . . . .                 | 28        |
| 2.5                                                  | Critical Constants from Van der Waals Equation . . . . .                     | 28        |
| 2.6                                                  | Hydrogen Bonding in Detail . . . . .                                         | 29        |
| 2.7                                                  | Corresponding States and Reduced Variables . . . . .                         | 29        |
| 2.8                                                  | Boyle Temperature . . . . .                                                  | 29        |
| 2.9                                                  | Joule-Thomson Effect . . . . .                                               | 30        |
| 2.10                                                 | Potential Energy Surface in Chemical Reactions . . . . .                     | 30        |
| 2.11                                                 | Applications of Intermolecular Forces in Engineering Chemistry . . . . .     | 30        |
| <b>Module 4: Free Energy and Chemical Equilibria</b> |                                                                              | <b>34</b> |
| <b>1</b>                                             | <b>Free Energy and Chemical Equilibria . . . . .</b>                         | <b>34</b> |
| 1.1                                                  | First and Second Laws of Thermodynamics . . . . .                            | 34        |
| 1.2                                                  | Entropy and Free Energy Estimation . . . . .                                 | 34        |
| 1.3                                                  | Free Energy and Equilibrium Constant . . . . .                               | 34        |
| 1.4                                                  | Cell Potential and Nernst Equation . . . . .                                 | 35        |
| 1.5                                                  | Acid-Base, Redox and Solubility Equilibria . . . . .                         | 35        |
| 1.6                                                  | Water Chemistry, Corrosion and Ellingham Diagram . . . . .                   | 35        |
| <b>2</b>                                             | <b>Derivations, Numericals and Engineering Applications . . . . .</b>        | <b>36</b> |
| 2.1                                                  | Deriving the Equilibrium Relation . . . . .                                  | 36        |
| 2.2                                                  | Entropy Change for Common Processes . . . . .                                | 37        |
| 2.3                                                  | Nernst Equation Derivation from Free Energy . . . . .                        | 37        |
| 2.4                                                  | Water Hardness and Boiler Problems . . . . .                                 | 37        |
| 2.5                                                  | Electrochemical Corrosion Mechanism . . . . .                                | 38        |
| 2.6                                                  | Ellingham Diagram: Reduction Rule . . . . .                                  | 38        |
| 2.7                                                  | Temperature Dependence of Spontaneity . . . . .                              | 38        |
| 2.8                                                  | pH, Acid Dissociation and Buffer Idea . . . . .                              | 39        |
| 2.9                                                  | Solubility Product and Common Ion Effect . . . . .                           | 39        |
| 2.10                                                 | Corrosion Prevention Methods . . . . .                                       | 40        |
| <b>Module 5: Periodic Properties</b>                 |                                                                              | <b>43</b> |
| <b>1</b>                                             | <b>Periodic Properties . . . . .</b>                                         | <b>43</b> |
| 1.1                                                  | Effective Nuclear Charge and Shielding . . . . .                             | 43        |
| 1.2                                                  | Penetration and Orbital Energy . . . . .                                     | 43        |

|                                  |                                                                                   |           |
|----------------------------------|-----------------------------------------------------------------------------------|-----------|
| 1.3                              | Atomic Size, Ionization Energy, Electron Affinity and Electronegativity . . . . . | 44        |
| 1.4                              | Polarizability, Oxidation States and Coordination . . . . .                       | 44        |
| 1.5                              | Hard and Soft Acids and Bases . . . . .                                           | 44        |
| 1.6                              | Molecular Geometries . . . . .                                                    | 45        |
| <b>2</b>                         | <b>Detailed Periodic Trend Analysis . . . . .</b>                                 | <b>45</b> |
| 2.1                              | Shielding Order and Slater-Type Reasoning . . . . .                               | 45        |
| 2.2                              | Successive Ionization Energies . . . . .                                          | 45        |
| 2.3                              | Diagonal Relationship . . . . .                                                   | 46        |
| 2.4                              | Oxidation States and Inert Pair Effect . . . . .                                  | 46        |
| 2.5                              | HSAB Applications . . . . .                                                       | 46        |
| 2.6                              | VSEPR Shapes and Lone Pair Effect . . . . .                                       | 47        |
| 2.7                              | Atomic and Ionic Radius: Detailed Trend . . . . .                                 | 47        |
| 2.8                              | Electron Affinity and Exceptions . . . . .                                        | 48        |
| 2.9                              | Electronegativity Scales . . . . .                                                | 48        |
| 2.10                             | Polarization and Fajan's Rules . . . . .                                          | 48        |
| 2.11                             | Coordination Number and Geometry in Complexes . . . . .                           | 48        |
| 2.12                             | Common Exceptions and Exam Traps . . . . .                                        | 49        |
| <b>Module 6: Stereochemistry</b> |                                                                                   | <b>52</b> |
| <b>1</b>                         | <b>Stereochemistry . . . . .</b>                                                  | <b>52</b> |
| 1.1                              | Representations of Three-Dimensional Structures . . . . .                         | 52        |
| 1.2                              | Structural Isomers and Stereoisomers . . . . .                                    | 52        |
| 1.3                              | Chirality, Enantiomers and Diastereomers . . . . .                                | 53        |
| 1.4                              | Optical Activity and Absolute Configuration . . . . .                             | 53        |
| 1.5                              | Elements of Symmetry . . . . .                                                    | 53        |
| 1.6                              | Conformational Analysis . . . . .                                                 | 54        |
| 1.7                              | Isomerism in Transition Metal Compounds . . . . .                                 | 54        |
| <b>2</b>                         | <b>Detailed Stereochemical Analysis . . . . .</b>                                 | <b>55</b> |
| 2.1                              | Fischer Projection Rules . . . . .                                                | 55        |
| 2.2                              | Assigning R and S Configuration . . . . .                                         | 55        |
| 2.3                              | Geometrical Isomerism: E/Z and Cis/Trans . . . . .                                | 56        |
| 2.4                              | Meso Compounds . . . . .                                                          | 56        |
| 2.5                              | Conformational Analysis of Butane . . . . .                                       | 56        |
| 2.6                              | Coordination Compound Stereoisomerism . . . . .                                   | 56        |
| 2.7                              | Number of Stereoisomers . . . . .                                                 | 57        |
| 2.8                              | Resolution of Racemic Mixtures . . . . .                                          | 57        |
| 2.9                              | Optical Purity and Enantiomeric Excess . . . . .                                  | 58        |
| 2.10                             | Sequence Rules for Double Bonds: E/Z Worked Method . . . . .                      | 58        |
| 2.11                             | Energy Profile of Butane Rotation . . . . .                                       | 58        |
| 2.12                             | Stereochemistry in Biological and Drug Action . . . . .                           | 58        |

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>Module 7: Organic Reactions and Drug Molecule Synthesis</b>  | <b>62</b> |
| <b>1 Organic Reactions and Drug Molecule Synthesis</b>          | <b>62</b> |
| 1.1 Substitution Reactions                                      | 62        |
| 1.2 Addition and Elimination Reactions                          | 62        |
| 1.3 Oxidation, Reduction, Cyclization and Ring Opening          | 63        |
| 1.4 Synthesis of a Common Drug Molecule: Aspirin                | 63        |
| 1.5 Synthesis of a Common Drug Molecule: Paracetamol            | 63        |
| 1.6 Mechanistic Exam Language                                   | 64        |
| <b>2 Detailed Organic Reaction Notes and Synthesis Planning</b> | <b>65</b> |
| 2.1 Reaction Intermediates                                      | 65        |
| 2.2 Detailed Comparison of $S_N1$ and $S_N2$                    | 65        |
| 2.3 Electrophilic Addition to Alkenes                           | 65        |
| 2.4 Elimination: E1 and E2                                      | 66        |
| 2.5 Oxidation and Reduction Reagents                            | 66        |
| 2.6 Cyclization and Ring Opening                                | 66        |
| 2.7 Aspirin Synthesis: Mechanism and Purification               | 67        |
| 2.8 Reaction Map for Basic Organic Transformations              | 67        |
| 2.9 Curly-Arrow Logic in Words                                  | 68        |
| 2.10 Zaitsev and Hofmann Elimination                            | 68        |
| 2.11 Addition to Carbonyl Compounds                             | 68        |
| 2.12 Green Chemistry Note for Drug Synthesis                    | 68        |
| 2.13 Tests for Functional Groups                                | 69        |

# Module 1

## Atomic and Molecular Structure

### 1. Atomic and Molecular Structure

**Atomic and molecular structure** explains the arrangement of electrons in atoms and molecules, the origin of bonding, and the connection between energy levels and observable properties.

#### 1.1 Schrodinger Wave Equation

In quantum chemistry, an electron is described by a wave function  $\psi$ . The square of the wave function gives probability density, so  $|\psi|^2 d\tau$  is the probability of finding the electron in a small volume element  $d\tau$ .

##### Time-Independent Schrodinger Equation

$$\hat{H}\psi = E\psi$$

where  $\hat{H}$  is the Hamiltonian operator,  $\psi$  is the wave function, and  $E$  is the permitted energy. For a one-dimensional particle,

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi$$

#### 1.2 Particle in a One-Dimensional Box

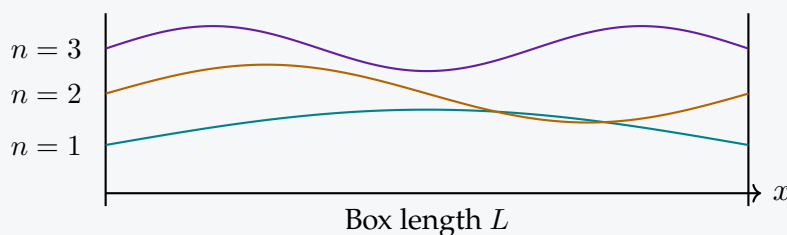
For a particle confined between  $x = 0$  and  $x = L$ , the potential energy is zero inside the box and infinite outside it. Boundary conditions are:

$$\psi(0) = 0, \quad \psi(L) = 0$$

The allowed wave functions and energies are:

$$\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \quad E_n = \frac{n^2 h^2}{8mL^2}, \quad n = 1, 2, 3, \dots$$

### Particle in a Box: Quantized Standing Waves



### 1.3 Molecular Orbitals of Diatomic Molecules

Molecular orbital (MO) theory states that atomic orbitals combine to form molecular orbitals spread over the whole molecule. Constructive overlap gives bonding MOs and destructive overlap gives antibonding MOs.

#### Bond Order

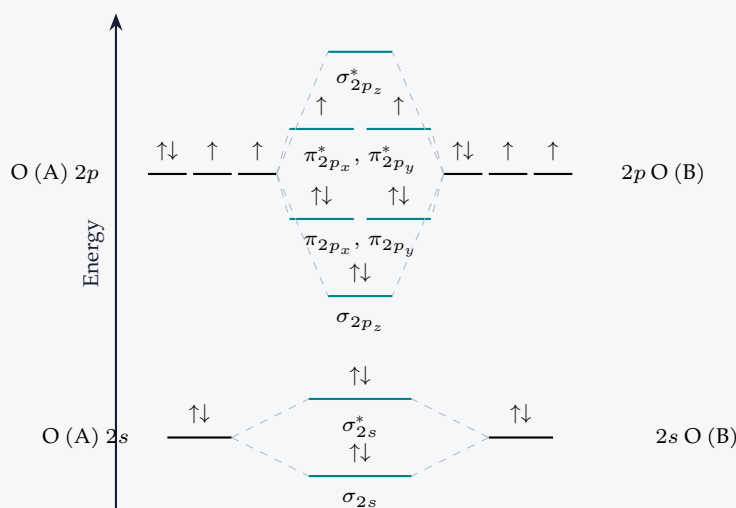
$$\text{Bond order} = \frac{N_b - N_a}{2}$$

where  $N_b$  is the number of electrons in bonding MOs and  $N_a$  is the number of electrons in antibonding MOs. Higher bond order means shorter, stronger bond.

#### Important examples:

| Molecule       | Configuration idea             | Bond order | Magnetism   |
|----------------|--------------------------------|------------|-------------|
| H <sub>2</sub> | Two electrons in $\sigma_{1s}$ | 1          | Diamagnetic |

### Molecular Orbital Energy Level Diagram of Oxygen (O<sub>2</sub>)

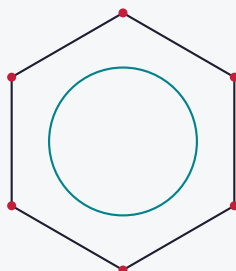


### 1.4 Pi-Molecular Orbitals and Aromaticity

Pi-electrons in conjugated systems are delocalized. Benzene has six pi-electrons distributed over six carbon atoms, making it unusually stable.

**Aromatic compound:** A cyclic, planar, fully conjugated molecule having  $(4n + 2)$  pi-electrons, where  $n = 0, 1, 2, \dots$

#### Benzene Pi-Electron Delocalization



6 delocalized pi-electrons satisfy Huckel rule

### 1.5 Pi-Molecular Orbitals of 1,3-Butadiene

1,3-butadiene ( $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$ ) has a conjugated system of four  $p$  orbitals containing four  $\pi$  electrons. These combine to form four  $\pi$  molecular orbitals ( $\psi_1, \psi_2, \psi_3, \psi_4$ ) with increasing energy and nodes.

- **Nodal Properties:** The number of nodes in orbital  $\psi_k$  is equal to  $(k - 1)$ . A node is a position where the wave function changes sign and electron probability is zero.
- **HOMO/LUMO:** In the ground state, the four  $\pi$  electrons fill the two lowest-energy orbitals ( $\psi_1, \psi_2$ ). Thus,  $\psi_2$  is the **Highest Occupied Molecular Orbital (HOMO)** and  $\psi_3$  is the **Lowest Unoccupied Molecular Orbital (LUMO)**.

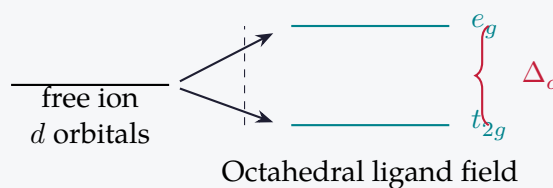
#### Pi-Molecular Orbitals of 1,3-Butadiene



### 1.6 Crystal Field Theory

In transition-metal complexes, surrounding ligands split the five degenerate  $d$  orbitals into different energy groups. The splitting controls colour, magnetic behaviour, and stability.

## Octahedral Crystal Field Splitting



## 1.7 Band Structure and Doping

In solids, many atomic orbitals combine to form bands. Conductivity depends on the band gap  $E_g$  between valence band and conduction band.

| Material      | Band gap                            | Electrical behaviour                     |
|---------------|-------------------------------------|------------------------------------------|
| Conductor     | Overlapping bands or very small gap | Electrons move easily                    |
| Semiconductor | Moderate gap, about 1–3 eV          | Conductivity increases by heat or doping |
| Insulator     | Large gap, usually > 5 eV           | Very poor conductor                      |

## Exam Points

- Derive  $E_n = n^2 h^2 / (8mL^2)$  using boundary conditions.
- For MO questions, always calculate bond order and then state magnetic nature.
- For CFT, mention splitting pattern,  $\Delta_o$ , high spin or low spin, and magnetic moment.
- For band theory, draw valence band, conduction band, and band gap.

## 2. Exam-Oriented Derivations and Applications

## 2.1 Derivation of Particle in a Box Energy

For a one-dimensional box, potential energy inside the box is zero. Therefore the Schrodinger equation becomes:

$$-\frac{h^2}{8\pi^2 m} \frac{d^2 \psi}{dx^2} = E\psi$$

Rearranging,

$$\frac{d^2 \psi}{dx^2} + \frac{8\pi^2 m E}{h^2} \psi = 0$$

Let

$$k^2 = \frac{8\pi^2 m E}{h^2}$$

Then the differential equation becomes:

$$\frac{d^2 \psi}{dx^2} + k^2 \psi = 0$$

Its general solution is:

$$\psi = A \sin kx + B \cos kx$$

Boundary condition at  $x = 0$  gives  $B = 0$ . Boundary condition at  $x = L$  gives:

$$A \sin kL = 0$$

For a non-zero wave function:

$$kL = n\pi, \quad k = \frac{n\pi}{L}$$

Substituting in  $k^2 = 8\pi^2mE/h^2$ :

$$E_n = \frac{n^2 h^2}{8mL^2}$$

#### Solved Example: Energy Level Ratio

**Problem:** Find the ratio of the first three energy levels for a particle in a box.

**Solution:** Since  $E_n \propto n^2$ ,

$$E_1 : E_2 : E_3 = 1^2 : 2^2 : 3^2 = 1 : 4 : 9$$

**Exam note:** Energy gaps are not equal; they increase with  $n$ .

## 2.2 MO Filling Order and Magnetic Behaviour

For second-period diatomic molecules, MO ordering changes after  $N_2$  because  $s$ - $p$  mixing becomes less important. This is a common exam trap.

| Molecules        | MO order to remember                                                                | Important result                                    |
|------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------|
| $B_2, C_2, N_2$  | $\sigma_{2s} < \sigma_{2s}^* < \pi_{2p} < \sigma_{2p} < \pi_{2p}^* < \sigma_{2p}^*$ | $B_2$ is paramagnetic; $N_2$ has bond order 3       |
| $O_2, F_2, Ne_2$ | $\sigma_{2s} < \sigma_{2s}^* < \sigma_{2p} < \pi_{2p} < \pi_{2p}^* < \sigma_{2p}^*$ | $O_2$ is paramagnetic due to two unpaired electrons |

#### Solved Example: Bond Order of Oxygen

For  $O_2$ , total valence electrons are 12.

$$N_b = 8, \quad N_a = 4$$

$$\text{Bond order} = \frac{8 - 4}{2} = 2$$

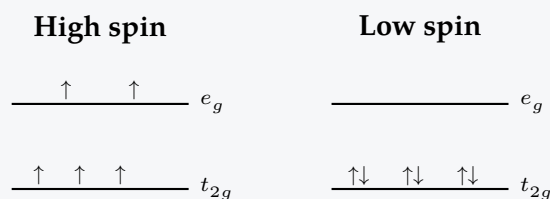
Two electrons remain unpaired in  $\pi_{2p}^*$  orbitals, so oxygen is **paramagnetic**.

## 2.3 Crystal Field Splitting: High Spin and Low Spin

In octahedral complexes, electrons first occupy lower  $t_{2g}$  orbitals. Whether pairing occurs depends on the relative values of crystal field splitting energy  $\Delta_o$  and pairing energy  $P$ .

| Case              | Condition      | Result                                                                                                     |
|-------------------|----------------|------------------------------------------------------------------------------------------------------------|
| High spin complex | $\Delta_o < P$ | Electrons avoid pairing; more unpaired electrons; usually weak-field ligands                               |
| Low spin complex  | $\Delta_o > P$ | Electrons pair in $t_{2g}$ before occupying $e_g$ ; fewer unpaired electrons; usually strong-field ligands |

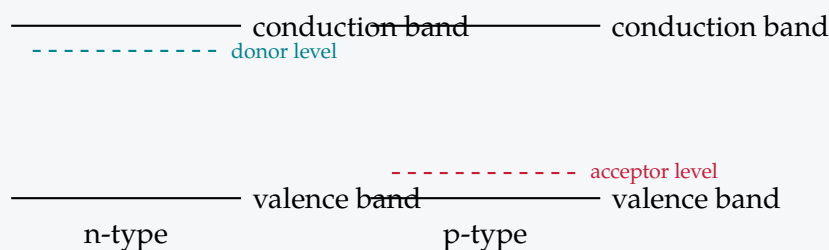
#### High Spin and Low Spin Choice for Octahedral Complexes



## 2.4 n-Type and p-Type Semiconductors

Doping adds controlled impurities to a semiconductor. Pentavalent dopants such as P or As introduce donor levels and produce n-type semiconductors. Trivalent dopants such as B or Al introduce acceptor levels and produce p-type semiconductors.

#### Doping Levels in Semiconductors



#### How to Write 10-Mark Answers in This Module

1. Start with a definition in 2–3 lines.
2. Draw the required energy-level or orbital diagram.
3. Write the governing formula with symbol meanings.
4. Add one example such as  $O_2$  paramagnetism, benzene aromaticity, or n-type doping.
5. End with one application or exam conclusion.

## 2.5 Important Long-Answer: Aromaticity of Benzene

Benzene is the standard example of aromatic stability. A complete answer should not only state Huckel rule, but also connect structure, pi-electron count, planarity, and equal bond lengths.

1. Benzene is cyclic and planar, so all six carbon atoms can overlap through unhybridized  $p$

orbitals.

- Each carbon contributes one pi-electron, giving a total of six pi-electrons.
- Huckel rule requires  $(4n + 2)$  pi-electrons. For benzene:

$$4n + 2 = 6 \Rightarrow n = 1$$

- The pi-electrons are delocalized above and below the ring.
- Delocalization makes all C–C bonds equivalent and gives extra stability.

#### Exam Comparison: Aromatic, Anti-Aromatic and Non-Aromatic

| Type                 | Condition                      | Pi-electrons        | Example idea                    |
|----------------------|--------------------------------|---------------------|---------------------------------|
| <b>Aromatic</b>      | Cyclic, planar, conjugated     | $(4n + 2)$          | Benzene, cyclopentadienyl anion |
| <b>Anti-aromatic</b> | Cyclic, planar, conjugated     | $4n$                | Cyclobutadiene type systems     |
| <b>Non-aromatic</b>  | Fails planarity or conjugation | Rule not applicable | Non-planar or saturated ring    |

## 2.6 Important Long-Answer: Crystal Field Theory and Magnetism

Crystal field theory explains colour and magnetic properties of transition metal complexes by ligand-induced splitting of  $d$  orbitals.

### Crystal Field Stabilization Energy

For an octahedral complex:

$$\text{CFSE} = (-0.4x + 0.6y)\Delta_o + pP$$

where  $x$  is the number of electrons in  $t_{2g}$  orbitals,  $y$  is the number of electrons in  $e_g$  orbitals, and  $pP$  is the pairing energy contribution.

### Solved Example: Octahedral $d^3$ Configuration

For a  $d^3$  octahedral ion, electrons occupy  $t_{2g}$  orbitals singly:

$$t_{2g}^3 e_g^0$$

$$\text{CFSE} = 3(-0.4\Delta_o) = -1.2\Delta_o$$

There are three unpaired electrons, so the spin-only magnetic moment is:

$$\mu = \sqrt{3(3 + 2)} = \sqrt{15} = 3.87 \text{ BM}$$

## 2.7 Band Theory: Exam Numericals and Concepts

Band gap decides electrical behaviour. If the gap is small, thermal energy can excite electrons to the conduction band. Doping reduces the effective energy needed for charge conduction.

| Question type                     | What to write                                        | Common conclusion                          |
|-----------------------------------|------------------------------------------------------|--------------------------------------------|
| <b>Conductor vs semiconductor</b> | Draw overlapping band and small band gap             | Conductor has many free electrons          |
| <b>Intrinsic semiconductor</b>    | Pure semiconductor with thermally generated carriers | Electron and hole concentrations are equal |
| <b>n-type semiconductor</b>       | Donor level near conduction band                     | Electrons are majority carriers            |
| <b>p-type semiconductor</b>       | Acceptor level near valence band                     | Holes are majority carriers                |

### Previous-Year Style Questions

1. Derive the energy expression for a particle in a one-dimensional box and explain why  $n = 0$  is not allowed.
2. Draw the MO diagram of  $O_2$  and explain its paramagnetic nature.
3. Explain aromaticity of benzene using Huckel rule.
4. Explain high-spin and low-spin octahedral complexes with suitable examples.
5. Compare conductors, semiconductors and insulators using band theory.

## Quick Revision — Module 1

Atomic and Molecular Structure | BS-CH101

### Key Formulas at a Glance

| Formula / Rule                                      | Meaning                                               |
|-----------------------------------------------------|-------------------------------------------------------|
| $\hat{H}\psi = E\psi$                               | Schrodinger eigenvalue equation for allowed energies. |
| $E_n = \frac{n^2 h^2}{8mL^2}$                       | Energy of a particle in a one-dimensional box.        |
| $\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$ | Normalized wave function for a particle in a box.     |
| <b>Bond order</b> = $\frac{N_b - N_a}{2}$           | Predicts bond strength and stability in MO theory.    |
| $\mu = \sqrt{n(n+2)} \text{ BM}$                    | Spin-only magnetic moment for $n$ unpaired electrons. |
| $4n + 2$                                            | Huckel rule for aromatic pi-electron count.           |

### One-Line Definitions

- **Wave function:** Mathematical function whose square gives probability density.
- **Node:** Point or surface where  $\psi = 0$ .
- **Bonding MO:** Lower-energy MO formed by constructive overlap.
- **Antibonding MO:** Higher-energy MO formed by destructive overlap.
- **Aromaticity:** Extra stability due to cyclic delocalization of  $(4n + 2)$  pi-electrons.
- **Crystal field splitting:** Separation of metal  $d$  orbitals in a ligand field.
- **Band gap:** Energy gap between valence and conduction bands.

### Frequently Asked / Viva Questions

**Q1. State the Schrodinger equation.**

The time-independent form is  $\hat{H}\psi = E\psi$ , where  $\hat{H}$  is the energy operator.

**Q2. Why is energy quantized for a particle in a box?**

Only wave functions satisfying  $\psi(0) = 0$  and  $\psi(L) = 0$  are allowed, giving discrete  $n$  values.

**Q3. What is the zero-point energy?**

The minimum energy of a confined particle,  $E_1 = h^2/(8mL^2)$ .

**Q4. Define bond order.**

Bond order is  $(N_b - N_a)/2$ ; it measures net bonding in a molecule.

## Frequently Asked / Viva Questions (contd.)

**Q5. Why is oxygen paramagnetic?**

O<sub>2</sub> has two unpaired electrons in antibonding  $\pi^*$  orbitals.

**Q6. What is Huckel rule?**

A cyclic planar conjugated molecule is aromatic if it has  $(4n + 2)$  pi-electrons.

**Q7. Why is benzene stable?**

Six pi-electrons are delocalized over the ring, satisfying Huckel rule with  $n = 1$ .

**Q8. What is crystal field splitting?**

Splitting of metal  $d$  orbitals into groups of different energy by ligands.

**Q9. What is the role of doping in semiconductors?**

Doping introduces donor or acceptor levels and increases carrier concentration.

**Q10. How are conductors and insulators distinguished by band theory?**

Conductors have overlapping or nearly touching bands, whereas insulators have a large band gap.

## Summary Reference Table

| Topic           | Key idea                                                 | Exam output                                    |
|-----------------|----------------------------------------------------------|------------------------------------------------|
| Particle in box | Boundary conditions give discrete energies               | Derivation plus energy-level diagram           |
| MO theory       | Atomic orbitals combine into bonding and antibonding MOs | Bond order and magnetism                       |
| CFT             | Ligands split $d$ orbitals                               | Splitting diagram and unpaired electrons       |
| Band theory     | Bands decide electrical conductivity                     | Conductor, semiconductor, insulator comparison |

# Module 2

## Spectroscopic Techniques and Applications

### 1. Spectroscopic Techniques and Applications

**Spectroscopy** is the study of interaction between electromagnetic radiation and matter. It is used to identify molecules, determine structure, and study energy-level transitions.

#### 1.1 Basic Principle and Selection Rules

When radiation of frequency  $\nu$  is absorbed or emitted, a molecule moves between two energy levels.

##### Energy of Radiation

$$\Delta E = h\nu = \frac{hc}{\lambda}$$

where  $h$  is Planck's constant,  $\nu$  is frequency,  $c$  is speed of light, and  $\lambda$  is wavelength.

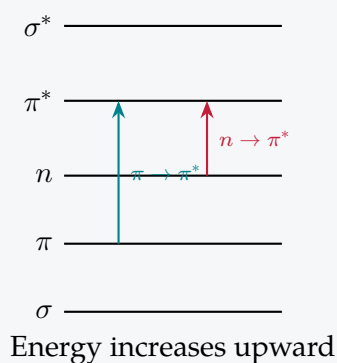
A transition is allowed only when it satisfies the required selection rule and has a changing electric or magnetic moment.

| Technique   | Energy change            | Common selection rule                             |
|-------------|--------------------------|---------------------------------------------------|
| Rotational  | Molecular rotation       | $\Delta J = \pm 1$                                |
| Vibrational | Bond vibration           | $\Delta v = \pm 1$ for fundamental transition     |
| Electronic  | Electron promotion       | Spin and symmetry allowed transitions are intense |
| NMR         | Nuclear spin orientation | Nuclei must have non-zero spin                    |

#### 1.2 Electronic Spectroscopy

Electronic spectroscopy uses UV-visible radiation. It involves transitions such as  $\sigma \rightarrow \sigma^*$ ,  $n \rightarrow \sigma^*$ ,  $\pi \rightarrow \pi^*$ , and  $n \rightarrow \pi^*$ .

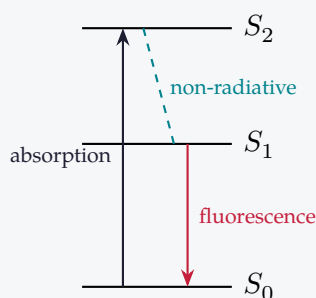
## Common Electronic Transitions



## 1.3 Fluorescence and Medical Applications

Fluorescence occurs when a molecule absorbs light and quickly emits lower-energy light. It is useful in diagnostics, bio-imaging, and detection of trace biomolecules.

## Jablonski Diagram for Fluorescence



## 1.4 Vibrational and Rotational Spectroscopy

IR spectroscopy studies molecular vibrations. A vibration is IR active if it changes dipole moment. Rotational spectroscopy is observed mainly for gas-phase polar molecules.

## Vibrational Frequency of a Diatomic Molecule

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}, \quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

where  $k$  is force constant and  $\mu$  is reduced mass.

## 1.5 NMR, MRI, Surface Characterisation, Diffraction

NMR detects nuclei in a magnetic field. MRI is a medical imaging application of NMR, mainly using hydrogen nuclei in water and fat. Surface characterisation methods such as XPS, SEM, and AFM help study surface composition and morphology. Diffraction and scattering reveal crystal structure, particle size, and defects.

**Bragg Equation**

$$n\lambda = 2d \sin \theta$$

Here  $d$  is interplanar spacing and  $\theta$  is the Bragg angle.

**Exam Points**

- Mention the energy region: microwave for rotational, IR for vibrational, UV-visible for electronic.
- In fluorescence answers, draw the Jablonski diagram.
- For NMR/MRI, write the need for non-zero nuclear spin.
- For X-ray diffraction, Bragg equation is compulsory.

## 2. Detailed Spectroscopy Notes for Exam Answers

### 2.1 Electromagnetic Spectrum and Type of Transition

Each spectroscopic method works in a particular energy region. In exam answers, first identify the radiation region, then the molecular transition.

| Region         | Approximate wavelength / frequency | Transition observed             |
|----------------|------------------------------------|---------------------------------|
| Radiofrequency | MHz range                          | Nuclear spin transitions in NMR |
| Microwave      | 1 mm to 1 m                        | Rotational transitions          |
| Infrared       | 4000–400 $\text{cm}^{-1}$          | Vibrational transitions         |
| Visible / UV   | 800–200 nm                         | Electronic transitions          |
| X-ray          | Very short wavelength              | Diffraction by crystal planes   |

### 2.2 Beer-Lambert Law in Electronic Spectroscopy

UV-visible spectroscopy is often used for concentration measurement. If monochromatic light passes through a solution:

$$A = \log \frac{I_0}{I}$$

where  $I_0$  is incident intensity and  $I$  is transmitted intensity.

**Beer-Lambert Law**

$$A = \epsilon cl$$

where  $A$  is absorbance,  $\epsilon$  is molar absorptivity in  $\text{L mol}^{-1} \text{cm}^{-1}$ ,  $c$  is concentration in  $\text{mol L}^{-1}$ , and  $l$  is path length in cm.

**Solved Example: Absorbance**

If  $\varepsilon = 2.0 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1}$ ,  $c = 1.0 \times 10^{-5} \text{ mol L}^{-1}$ , and  $l = 1 \text{ cm}$ :

$$A = \varepsilon cl = (2.0 \times 10^4)(1.0 \times 10^{-5})(1) = 0.20$$

So the solution has absorbance 0.20.

**2.3 IR Spectroscopy: Functional Group Region**

IR spectra are very useful for identifying functional groups. The fingerprint region helps confirm identity, while the functional group region gives quick structural clues.

| Bond / group | Typical range              | Exam identification               |
|--------------|----------------------------|-----------------------------------|
| O–H stretch  | 3200–3600 $\text{cm}^{-1}$ | Broad band for alcohols and acids |
| N–H stretch  | 3300–3500 $\text{cm}^{-1}$ | Amine or amide indication         |
| C=O stretch  | 1650–1800 $\text{cm}^{-1}$ | Strong sharp carbonyl peak        |
| C=C stretch  | 1600–1680 $\text{cm}^{-1}$ | Alkene or aromatic ring           |
| C–O stretch  | 1000–1300 $\text{cm}^{-1}$ | Alcohol, ether, ester             |

**2.4 NMR Chemical Shift and Splitting**

In proton NMR, chemical shift tells the electronic environment. Spin-spin splitting tells the number of neighbouring equivalent protons.

**NMR Splitting Rule**

If a proton has  $n$  equivalent neighbouring protons, its signal is split into:

$$n + 1 \text{ peaks}$$

The relative intensities approximately follow Pascal's triangle.

| Proton type               | Typical chemical shift | Reason                       |
|---------------------------|------------------------|------------------------------|
| Alkyl R – CH <sub>3</sub> | 0.8–1.5 ppm            | Shielded environment         |
| Near electronegative atom | 3–4.5 ppm              | Deshielding by O, N, halogen |
| Aromatic proton           | 6.5–8 ppm              | Ring current effect          |
| Aldehydic proton          | 9–10 ppm               | Strongly deshielded          |

**2.5 Bragg Law Derivation Idea**

Constructive interference occurs when the path difference between X-rays reflected from successive crystal planes is an integral multiple of wavelength.

$$\text{Path difference} = AB + BC = 2d \sin \theta$$

For constructive interference:

$$n\lambda = 2d \sin \theta$$

This equation is used to calculate interplanar spacing  $d$  from diffraction angle.

#### Solved Example: Crystal Plane Spacing

For first-order reflection,  $\lambda = 1.54 \text{ \AA}$  and  $\theta = 30^\circ$ :

$$d = \frac{n\lambda}{2 \sin \theta} = \frac{1.54}{2 \sin 30^\circ}$$

$$d = \frac{1.54}{1} = 1.54 \text{ \AA}$$

#### Answer-Writing Pattern for Spectroscopy

1. State the radiation region and energy transition.
2. Give the selection rule or activity condition.
3. Draw the transition diagram if possible.
4. Mention one analytical or medical application.
5. For numerical questions, write units clearly.

## 2.6 Rotational Spectroscopy of a Rigid Diatomic Molecule

For a rigid rotor, rotational energy levels are quantized:

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1), \quad J = 0, 1, 2, \dots$$

where  $I = \mu r^2$  is moment of inertia,  $\mu$  is reduced mass, and  $r$  is bond length.

#### Rotational Spectral Lines

In wavenumber units:

$$\bar{\nu} = 2B(J+1), \quad B = \frac{h}{8\pi^2 I c}$$

Successive lines are equally spaced by:

$$\Delta \bar{\nu} = 2B$$

#### Solved Example: Rotational Line Spacing

If the rotational constant of a molecule is  $B = 5 \text{ cm}^{-1}$ , then adjacent rotational spectral lines are separated by:

$$\Delta \bar{\nu} = 2B = 10 \text{ cm}^{-1}$$

## 2.7 Vibrational Spectroscopy and Hooke's Law Model

A diatomic molecule behaves approximately like two masses connected by a spring. Stronger bonds have larger force constant  $k$  and therefore absorb at higher wavenumber.

| Factor                | Effect on frequency              | Reason                                  |
|-----------------------|----------------------------------|-----------------------------------------|
| Higher force constant | Frequency increases              | Bond is stiffer                         |
| Higher reduced mass   | Frequency decreases              | Heavier atoms vibrate more slowly       |
| Hydrogen bonding      | O–H/N–H peak becomes broad       | Different bond strengths in association |
| Conjugation near C=O  | C=O frequency decreases slightly | Partial single-bond character increases |

## 2.8 Electronic Spectra: Bathochromic and Hypsochromic Shifts

Electronic absorption changes when substituents or solvents change the energy gap between molecular orbitals.

| Term                | Meaning                          | Simple interpretation                           |
|---------------------|----------------------------------|-------------------------------------------------|
| Bathochromic shift  | Shift to longer wavelength       | Energy gap decreases; red shift                 |
| Hypsochromic shift  | Shift to shorter wavelength      | Energy gap increases; blue shift                |
| Hyperchromic effect | Increase in absorption intensity | More allowed transition or greater conjugation  |
| Hypochromic effect  | Decrease in absorption intensity | Less allowed transition or restricted structure |

## 2.9 Fluorescence Versus Phosphorescence

Fluorescence and phosphorescence are both emission processes, but their lifetimes and electronic states differ.

| Point       | Fluorescence                              | Phosphorescence                      |
|-------------|-------------------------------------------|--------------------------------------|
| Transition  | Singlet to singlet                        | Triplet to singlet                   |
| Lifetime    | Very short, about $10^{-9}$ – $10^{-7}$ s | Longer, milliseconds to seconds      |
| Spin rule   | Spin-allowed                              | Spin-forbidden, so slow              |
| Observation | Stops quickly when source is removed      | May continue after source is removed |

### Previous-Year Style Questions

1. Explain selection rules for rotational and vibrational spectroscopy.
2. Derive the expression for vibrational frequency of a diatomic molecule.
3. Explain fluorescence with Jablonski diagram and give medical applications.
4. State Beer-Lambert law and solve a concentration-based numerical.
5. Derive Bragg law and explain its use in crystal structure determination.

## 2.10 Short Notes on Surface Characterisation

Surface characterisation is used to study the outermost layers of solids, thin films, catalysts, sensors and biomaterials. These methods are important in engineering because surface properties often decide corrosion resistance, adhesion, catalytic activity and device performance.

| Technique | Information obtained                              | Engineering use                                    |
|-----------|---------------------------------------------------|----------------------------------------------------|
| SEM       | Surface morphology and particle shape             | Fracture surface, coating quality, microstructure  |
| AFM       | Nanometre-scale surface topography                | Thin films, MEMS surfaces, biomaterials            |
| XPS       | Surface elemental composition and oxidation state | Corrosion films, catalysts, semiconductor surfaces |
| XRD       | Crystal structure and interplanar spacing         | Phase identification and crystallinity             |

### Exam Answer: Why Surface Analysis Matters

Bulk composition may remain unchanged while the surface oxidizes, adsorbs impurities or forms a thin passive layer. Therefore surface techniques are needed to understand corrosion, catalysis, coating failure and sensor response.

## Quick Revision — Module 2

Spectroscopic Techniques and Applications | BS-CH101

### Key Formulas at a Glance

| Formula / Rule                              | Meaning                                            |
|---------------------------------------------|----------------------------------------------------|
| $\Delta E = h\nu = \frac{hc}{\lambda}$      | Photon energy absorbed or emitted in spectroscopy. |
| $\Delta J = \pm 1$                          | Rotational selection rule.                         |
| $\Delta v = \pm 1$                          | Fundamental vibrational transition rule.           |
| $\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$ | Diatomic vibrational frequency.                    |
| $\mu = \frac{m_1 m_2}{m_1 + m_2}$           | Reduced mass of a diatomic molecule.               |
| $n\lambda = 2d \sin \theta$                 | Bragg equation for diffraction.                    |

### One-Line Definitions

- **Spectroscopy:** Study of interaction of radiation with matter.
- **Selection rule:** Condition deciding whether a transition is allowed.
- **Chromophore:** Group responsible for UV-visible absorption.
- **Fluorescence:** Fast emission after electronic excitation.
- **IR active vibration:** Vibration causing a change in dipole moment.
- **Chemical shift:** NMR position relative to a reference.
- **Diffraction:** Constructive interference of waves scattered by ordered planes.

### Frequently Asked / Viva Questions

**Q1. What is spectroscopy?**

It is the study of absorption, emission, or scattering of radiation by matter.

**Q2. Write the relation between energy and wavelength.**

$$\Delta E = hc/\lambda.$$

**Q3. What is a selection rule?**

It is a rule that decides whether a spectroscopic transition is allowed.

**Q4. Why are rotational spectra shown by polar molecules?**

Rotation must produce an oscillating dipole moment.

**Q5. What is the condition for IR activity?**

## Frequently Asked / Viva Questions (contd.)

The vibration must change the dipole moment of the molecule.

**Q6. What is fluorescence?**

It is emission of light from an excited singlet state to the ground state.

**Q7. Give one medical use of fluorescence.**

Fluorescent probes are used in bio-imaging and disease diagnostics.

**Q8. What is NMR based on?**

Resonance absorption by nuclei with non-zero spin in a magnetic field.

**Q9. How is MRI related to NMR?**

MRI is an imaging technique based on NMR signals, mainly from hydrogen nuclei.

**Q10. State Bragg law.**

$$n\lambda = 2d \sin \theta.$$

## Summary Reference Table

| Method       | Radiation / signal      | Main application                   |
|--------------|-------------------------|------------------------------------|
| UV-visible   | Electronic transitions  | Conjugation and concentration      |
| Fluorescence | Emitted visible light   | Medical imaging and trace analysis |
| IR           | Vibrational transitions | Functional group identification    |
| NMR/MRI      | Nuclear spin resonance  | Structure and medical imaging      |
| XRD          | Diffraction X-rays      | Crystal structure                  |

# Module 3

## Intermolecular Forces and Potential Energy Surfaces

### 1. Intermolecular Forces and Potential Energy Surfaces

**Intermolecular forces** are attractive or repulsive forces between molecules or ions. They are weaker than covalent bonds but strongly affect boiling point, solubility, viscosity, and real-gas behaviour.

#### 1.1 Ionic, Dipole and Van der Waals Interactions

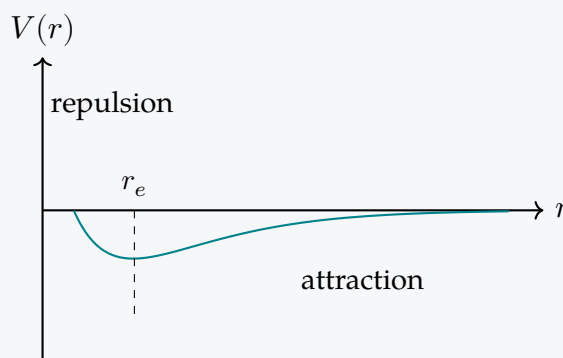
Ionic interactions are strongest among common non-covalent interactions. Dipole-dipole forces occur between polar molecules. London dispersion forces arise from temporary induced dipoles and exist in all molecules.

| Interaction       | Origin                                            | Example                         |
|-------------------|---------------------------------------------------|---------------------------------|
| Ion-ion           | Attraction between full charges                   | $\text{Na}^+$ and $\text{Cl}^-$ |
| Ion-dipole        | Ion attracts polar molecule                       | Hydration of $\text{Na}^+$      |
| Dipole-dipole     | Permanent dipoles align                           | HCl molecules                   |
| London dispersion | Instantaneous induced dipoles                     | Noble gases, hydrocarbons       |
| Hydrogen bonding  | H attached to F, O, or N interacts with lone pair | Water, alcohols                 |

#### 1.2 Potential Energy Curve

The potential energy between two particles depends on separation distance. At equilibrium distance  $r_e$ , attractive and repulsive forces balance and potential energy is minimum.

## Potential Energy Curve for Intermolecular Interaction



## 1.3 Equations of State for Real Gases

Ideal gas equation assumes no intermolecular force and zero molecular volume. Real gases deviate at high pressure and low temperature.

## Van der Waals Equation

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

where  $a$  corrects attractive forces and  $b$  corrects finite molecular volume.

## 1.4 Critical Phenomena

The critical temperature  $T_c$  is the temperature above which a gas cannot be liquefied by pressure alone. At the critical point, liquid and vapour become indistinguishable.

## Critical Constants for a Van der Waals Gas

$$V_c = 3nb, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

## Exam Points

- Compare intermolecular forces by strength and origin.
- Draw the potential energy curve and mark  $r_e$  and minimum energy.
- For real gases, explain physical meaning of constants  $a$  and  $b$ .
- Define critical temperature, pressure, and volume clearly.

## 2. Detailed Notes on Forces, Real Gases and Critical Behaviour

## 2.1 Origin and Strength of Intermolecular Forces

Intermolecular force strength depends on charge, polarity, polarizability, and molecular shape. Ionic interactions are strong because full charges are involved. Dispersion forces are weak individually but become important for large molecules with many electrons.

| Force             | Approximate strength trend | Property affected                   |
|-------------------|----------------------------|-------------------------------------|
| Ion-ion           | Very strong                | Melting point of ionic solids       |
| Ion-dipole        | Strong                     | Hydration and solubility of salts   |
| Hydrogen bond     | Moderate to strong         | Boiling point of water and alcohols |
| Dipole-dipole     | Moderate                   | Boiling point of polar molecules    |
| London dispersion | Weak to moderate           | Liquefaction of non-polar gases     |

## 2.2 Potential Energy and Force Relation

The force between two particles is related to the potential energy curve by:

$$F = -\frac{dV}{dr}$$

At the equilibrium separation  $r_e$ , potential energy is minimum, so:

$$\left(\frac{dV}{dr}\right)_{r=r_e} = 0, \quad F = 0$$

For  $r < r_e$ , repulsive force dominates. For  $r > r_e$ , attractive force dominates.

### Exam Interpretation of Potential Energy Curve

If two molecules are very far apart,  $V(r) \approx 0$ . As they approach, attraction lowers the potential energy. At  $r_e$ , the most stable separation is reached. If they come closer than  $r_e$ , electron-cloud repulsion sharply increases energy.

## 2.3 Lennard-Jones Potential

A common model for non-bonded interaction is:

$$V(r) = 4\varepsilon \left[ \left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6 \right]$$

The  $r^{-12}$  term represents short-range repulsion and the  $r^{-6}$  term represents London attraction.  $\varepsilon$  indicates well depth, i.e. strength of attraction.

| Term              | Physical meaning                                                |
|-------------------|-----------------------------------------------------------------|
| $(\sigma/r)^{12}$ | Repulsion due to electron-cloud overlap                         |
| $-(\sigma/r)^6$   | Attractive dispersion interaction                               |
| $\varepsilon$     | Depth of potential well; higher value means stronger attraction |
| $r_e$             | Separation at minimum potential energy                          |

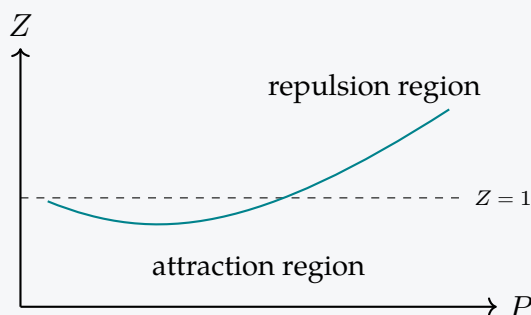
## 2.4 Compressibility Factor and Deviation from Ideality

Deviation from ideal gas behaviour is often measured using compressibility factor:

$$Z = \frac{PV}{nRT}$$

For an ideal gas,  $Z = 1$ . For a real gas,  $Z < 1$  indicates attraction-dominated behaviour and  $Z > 1$  indicates repulsion or finite-volume dominated behaviour.

### Compressibility Factor: Qualitative Behaviour



## 2.5 Critical Constants from Van der Waals Equation

For one mole:

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

At the critical point, the isotherm has a point of inflection:

$$\left(\frac{\partial P}{\partial V_m}\right)_T = 0, \quad \left(\frac{\partial^2 P}{\partial V_m^2}\right)_T = 0$$

Solving these conditions gives:

$$V_c = 3b, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

These relations are often asked as short derivation or formula-based questions.

### Solved Example: Critical Volume

If van der Waals constant  $b = 0.040 \text{ L mol}^{-1}$ , then:

$$V_c = 3b = 3(0.040) = 0.120 \text{ L mol}^{-1}$$

### How to Score in This Module

- Write force origin, example, and property affected for each intermolecular force.
- In potential energy questions, always mark attraction, repulsion,  $r_e$ , and energy minimum.
- For real gases, explain that  $a$  corrects pressure and  $b$  corrects volume.
- Use  $Z = PV/nRT$  to discuss positive and negative deviation.

## 2.6 Hydrogen Bonding in Detail

Hydrogen bonding occurs when hydrogen is covalently bonded to a highly electronegative atom such as F, O, or N and interacts with a lone pair on another electronegative atom.

| Type                                   | Meaning                                   | Example                       |
|----------------------------------------|-------------------------------------------|-------------------------------|
| <b>Intermolecular hydrogen bonding</b> | Hydrogen bond between different molecules | Water, alcohols, HF           |
| <b>Intramolecular hydrogen bonding</b> | Hydrogen bond within the same molecule    | o-nitrophenol type structures |
| <b>Strong hydrogen bonding</b>         | Shorter and more directional interaction  | HF, water network             |
| <b>Weak hydrogen bonding</b>           | Longer and less directional interaction   | C–H based weak interactions   |

### Exam Explanation: Boiling Point of Water

Water has a much higher boiling point than  $\text{H}_2\text{S}$  because oxygen is highly electronegative and forms extensive intermolecular hydrogen bonding. More heat is required to break this associated network before molecules escape into vapour phase.

## 2.7 Corresponding States and Reduced Variables

The law of corresponding states states that gases behave similarly when compared at the same reduced temperature and reduced pressure.

### Reduced Variables

$$T_r = \frac{T}{T_c}, \quad P_r = \frac{P}{P_c}, \quad V_r = \frac{V}{V_c}$$

where  $T_c$ ,  $P_c$ , and  $V_c$  are critical constants.

## 2.8 Boyle Temperature

At Boyle temperature, a real gas behaves nearly ideally over a considerable pressure range because attractive and repulsive deviations approximately cancel.

$$Z \approx 1$$

For a van der Waals gas, the Boyle temperature is approximately:

$$T_B = \frac{a}{Rb}$$

### Solved Example: Boyle Temperature

If  $a = 1.36 \text{ L}^2 \text{ atm mol}^{-2}$  and  $b = 0.0318 \text{ L mol}^{-1}$ :

$$T_B = \frac{a}{Rb} = \frac{1.36}{(0.0821)(0.0318)}$$

$$T_B \approx 521 \text{ K}$$

## 2.9 Joule-Thomson Effect

When a real gas expands through a porous plug from high pressure to low pressure without external work and without heat exchange, its temperature may change. This is called Joule-Thomson effect.

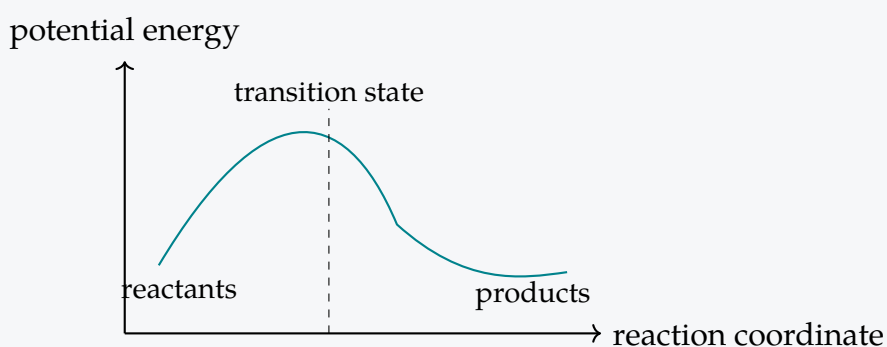
**Inversion temperature:** The temperature at which Joule-Thomson coefficient becomes zero. Below it, gas cools on expansion; above it, gas warms on expansion.

| Condition | Joule-Thomson coefficient | Observation            |
|-----------|---------------------------|------------------------|
| $T < T_i$ | Positive                  | Gas cools on expansion |
| $T = T_i$ | Zero                      | No temperature change  |
| $T > T_i$ | Negative                  | Gas warms on expansion |

## 2.10 Potential Energy Surface in Chemical Reactions

A potential energy surface shows how molecular energy changes with geometry. Reactants, products, intermediates, and transition states are represented as regions on the surface.

### Reaction Path on a Potential Energy Surface



### Previous-Year Style Questions

1. Compare ion-dipole, dipole-dipole, hydrogen bonding and dispersion forces.
2. Explain the potential energy curve for two interacting molecules.
3. Derive or state critical constants of a van der Waals gas.
4. Explain compressibility factor and deviation from ideal behaviour.
5. Write short notes on Joule-Thomson effect and inversion temperature.

## 2.11 Applications of Intermolecular Forces in Engineering Chemistry

Intermolecular forces are not only theoretical; they explain important material and chemical behaviour.

| Application                  | Force involved                               | Importance                                        |
|------------------------------|----------------------------------------------|---------------------------------------------------|
| Solubility of salts in water | Ion-dipole force                             | Hydration stabilizes ions in solution             |
| Polymer strength             | Dipole interaction and hydrogen bonding      | Controls toughness, flexibility and melting range |
| Adsorption on catalyst       | van der Waals and polar interactions         | Reactants attach to active surface sites          |
| Boiling point of liquids     | Hydrogen bonding and dispersion              | Stronger forces increase boiling point            |
| Protein structure            | Hydrogen bonding and hydrophobic interaction | Maintains secondary and tertiary structure        |

**Exam Answer: Like Dissolves Like**

Polar solvents dissolve ionic and polar solutes because ion-dipole or dipole-dipole interactions compensate for the energy needed to separate solute particles. Non-polar solutes dissolve better in non-polar solvents because dispersion forces are similar in both.

## Quick Revision — Module 3

Intermolecular Forces and Potential Energy Surfaces | BS-CH101

### Key Formulas at a Glance

| Formula / Rule                                    | Meaning                                         |
|---------------------------------------------------|-------------------------------------------------|
| $F = -\frac{dV}{dr}$                              | Force is negative gradient of potential energy. |
| $PV = nRT$                                        | Ideal gas equation.                             |
| $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ | Van der Waals real-gas equation.                |
| $V_c = 3nb$                                       | Critical volume for van der Waals gas.          |
| $P_c = \frac{a}{27b^2}$                           | Critical pressure for one mole form.            |
| $T_c = \frac{8a}{27Rb}$                           | Critical temperature for one mole form.         |

### One-Line Definitions

- **Intermolecular force:** Force acting between molecules or ions.
- **Dipole moment:** Product of charge and separation in a polar molecule.
- **Hydrogen bond:** Strong dipole interaction involving H bonded to F, O, or N.
- **Potential energy surface:** Energy variation with molecular geometry.
- **Real gas:** Gas that deviates from ideal behaviour due to volume and attraction.
- **Critical temperature:** Highest temperature at which gas can be liquefied by pressure.

### Frequently Asked / Viva Questions

**Q1. What are intermolecular forces?**

Forces of attraction or repulsion acting between separate molecules or ions.

**Q2. Arrange common forces in increasing strength.**

London dispersion < dipole-dipole < hydrogen bonding < ion-dipole < ion-ion.

**Q3. Why does boiling point increase with stronger intermolecular force?**

More energy is needed to separate molecules into vapour.

**Q4. What is equilibrium separation in a potential energy curve?**

It is the distance where potential energy is minimum and net force is zero.

**Q5. What causes repulsion at very small distance?**

Overlap of electron clouds and nuclear repulsion.

## Frequently Asked / Viva Questions (contd.)

**Q6. Why do real gases deviate from ideal behaviour?**

Molecules have finite volume and attractive forces.

**Q7. What is the role of  $a$  in van der Waals equation?**

It corrects pressure for intermolecular attraction.

**Q8. What is the role of  $b$  in van der Waals equation?**

It corrects volume for finite molecular size.

**Q9. Define critical point.**

The point at which liquid and vapour phases become indistinguishable.

**Q10. What is critical temperature?**

The highest temperature above which gas cannot be liquefied by pressure alone.

## Summary Reference Table

| Concept               | Meaning                             | Must write in exam                |
|-----------------------|-------------------------------------|-----------------------------------|
| Intermolecular forces | Non-covalent forces between species | Strength order and examples       |
| Potential curve       | Energy versus separation            | Minimum at $r_e$                  |
| Real gas              | Non-ideal gas                       | Van der Waals correction terms    |
| Critical point        | End point of liquid-vapour boundary | $T_c$ , $P_c$ , $V_c$ definitions |

# Module 4

## Free Energy and Chemical Equilibria

### 1. Free Energy and Chemical Equilibria

**Gibbs free energy** is the thermodynamic function that predicts spontaneity at constant temperature and pressure. A process is spontaneous when  $\Delta G < 0$ .

#### 1.1 First and Second Laws of Thermodynamics

The first law states conservation of energy. The second law states that total entropy of the universe increases for a spontaneous process.

##### Thermodynamic Functions

$$\Delta U = q + w, \quad \Delta H = \Delta U + \Delta(PV)$$

$$\Delta G = \Delta H - T\Delta S$$

At constant temperature and pressure,  $\Delta G < 0$  means spontaneous,  $\Delta G = 0$  means equilibrium, and  $\Delta G > 0$  means non-spontaneous.

#### 1.2 Entropy and Free Energy Estimation

Entropy measures dispersal of energy. For a reversible process,

$$\Delta S = \int \frac{\delta q_{\text{rev}}}{T}$$

Free energy change combines enthalpy and entropy and is useful because it directly predicts useful work and equilibrium direction.

#### 1.3 Free Energy and Equilibrium Constant

For a reaction at non-standard condition,

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium,  $\Delta G = 0$  and  $Q = K$ .

### Relation Between Free Energy and Equilibrium Constant

$$\Delta G^\circ = -RT \ln K = -2.303RT \log K$$

## 1.4 Cell Potential and Nernst Equation

Electrochemical cell work is linked with free energy.

### Electrochemical Free Energy

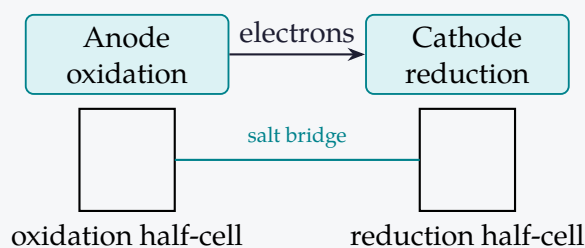
$$\Delta G = -nFE, \quad \Delta G^\circ = -nFE^\circ$$

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298 K:

$$E = E^\circ - \frac{0.0591}{n} \log Q$$

### Galvanic Cell and Electron Flow



## 1.5 Acid-Base, Redox and Solubility Equilibria

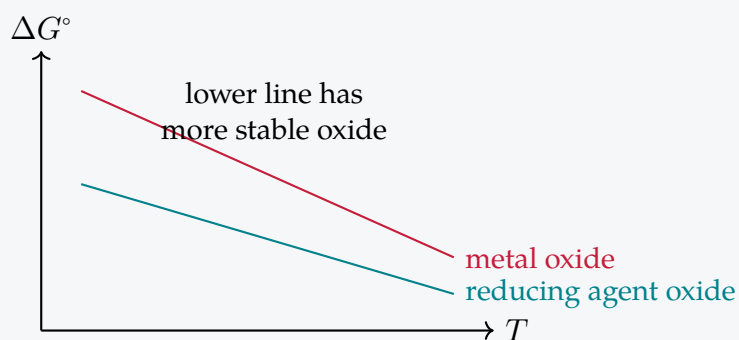
The same free-energy idea applies to acid-base ionization, redox reactions, and precipitation.

| Equilibrium | Constant     | Use                              |
|-------------|--------------|----------------------------------|
| Acid-base   | $K_a, K_b$   | pH and strength of acid/base     |
| Redox       | $E^\circ, K$ | Feasibility of electron transfer |
| Solubility  | $K_{sp}$     | Precipitation and ionic product  |

## 1.6 Water Chemistry, Corrosion and Ellingham Diagram

Water chemistry includes hardness, alkalinity, pH, and dissolved oxygen. Corrosion is an electrochemical oxidation of metal. Ellingham diagrams plot  $\Delta G^\circ$  versus temperature and help choose reducing agents in metallurgy.

## Ellingham Diagram Concept



## Exam Points

- Always state sign of  $\Delta G$  for spontaneity.
- For equilibrium questions, start from  $\Delta G = \Delta G^\circ + RT \ln Q$ .
- For electrochemistry, connect  $\Delta G$ ,  $E$ , and Nernst equation.
- For Ellingham diagrams, lower line means more stable oxide.

## 2. Derivations, Numericals and Engineering Applications

### 2.1 Deriving the Equilibrium Relation

For any reaction:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium, the reaction has no net driving force, so:

$$\Delta G = 0, \quad Q = K$$

Therefore:

$$0 = \Delta G^\circ + RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

Using  $\ln K = 2.303 \log K$ :

$$\Delta G^\circ = -2.303RT \log K$$

#### Solved Example: Equilibrium Constant from Free Energy

At 298 K, suppose  $\Delta G^\circ = -11.4 \text{ kJ mol}^{-1}$ .

$$\log K = \frac{-\Delta G^\circ}{2.303RT}$$

$$\log K = \frac{11400}{2.303(8.314)(298)} \approx 2.00$$

$$K \approx 10^2 = 100$$

Since  $K > 1$ , products are favoured.

## 2.2 Entropy Change for Common Processes

Entropy increases when randomness, volume, or number of gas molecules increases.

| Process             | Entropy change    | Reason                            |
|---------------------|-------------------|-----------------------------------|
| Melting of solid    | Positive          | Liquid has more molecular freedom |
| Vaporization        | Strongly positive | Gas has much greater randomness   |
| Compression of gas  | Negative          | Accessible volume decreases       |
| Dissolution of salt | Usually positive  | Ions disperse in solution         |
| Crystallization     | Negative          | Ordered solid forms               |

## 2.3 Nernst Equation Derivation from Free Energy

Electrical work from a cell is:

$$\Delta G = -nFE$$

At standard condition:

$$\Delta G^\circ = -nFE^\circ$$

Substitute both in:

$$\begin{aligned}\Delta G &= \Delta G^\circ + RT \ln Q \\ -nFE &= -nFE^\circ + RT \ln Q\end{aligned}$$

Dividing by  $-nF$ :

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298 K:

$$E = E^\circ - \frac{0.0591}{n} \log Q$$

### Solved Example: Cell Potential

For a cell with  $E^\circ = 1.10$  V,  $n = 2$ , and  $Q = 10$ :

$$\begin{aligned}E &= 1.10 - \frac{0.0591}{2} \log 10 \\ E &= 1.10 - 0.02955 = 1.07045 \text{ V}\end{aligned}$$

## 2.4 Water Hardness and Boiler Problems

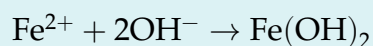
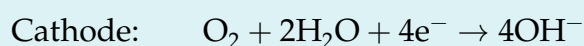
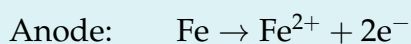
Hard water contains dissolved calcium and magnesium salts. Temporary hardness is due to bicarbonates; permanent hardness is due to chlorides and sulphates.

| Problem               | Cause                                    | Effect in engineering systems           |
|-----------------------|------------------------------------------|-----------------------------------------|
| Scale formation       | Deposition of salts like $\text{CaCO}_3$ | Poor heat transfer and tube overheating |
| Sludge formation      | Loose precipitates in boiler water       | Choking and reduced efficiency          |
| Priming               | Carryover of water droplets with steam   | Wet steam and turbine damage            |
| Foaming               | Stable bubbles due to impurities         | Irregular boiler operation              |
| Caustic embrittlement | Concentrated alkali in cracks            | Boiler metal failure                    |

## 2.5 Electrochemical Corrosion Mechanism

In iron corrosion, anodic and cathodic regions form on the metal surface.

### Rusting Half-Reactions



Further oxidation gives hydrated ferric oxide, commonly called rust.

## 2.6 Ellingham Diagram: Reduction Rule

In metallurgy, a metal oxide can be reduced by an element whose oxide line lies below it in the Ellingham diagram. This means the reducing agent forms a more stable oxide and makes the overall  $\Delta G^\circ$  negative.

### Answer-Writing Pattern for Ellingham Problems

1. Identify the oxide to be reduced.
2. Locate its  $\Delta G^\circ$  line at the given temperature.
3. Find a reducing agent whose oxide line is lower.
4. State that the net free energy change becomes negative.

## 2.7 Temperature Dependence of Spontaneity

The sign of  $\Delta G = \Delta H - T\Delta S$  depends on both enthalpy and entropy. This table is a common 5-mark answer.

| Case                           | Sign condition               | Spontaneity                         |
|--------------------------------|------------------------------|-------------------------------------|
| Exothermic, entropy increases  | $\Delta H < 0, \Delta S > 0$ | Spontaneous at all temperatures     |
| Endothermic, entropy decreases | $\Delta H > 0, \Delta S < 0$ | Non-spontaneous at all temperatures |
| Exothermic, entropy decreases  | $\Delta H < 0, \Delta S < 0$ | Spontaneous at low temperature      |
| Endothermic, entropy increases | $\Delta H > 0, \Delta S > 0$ | Spontaneous at high temperature     |

#### Solved Example: Temperature of Spontaneity

For a reaction,  $\Delta H = 40 \text{ kJ mol}^{-1}$  and  $\Delta S = 100 \text{ J mol}^{-1} \text{ K}^{-1}$ . Convert entropy:

$$\Delta S = 0.100 \text{ kJ mol}^{-1} \text{ K}^{-1}$$

At equilibrium,  $\Delta G = 0$ :

$$T = \frac{\Delta H}{\Delta S} = \frac{40}{0.100} = 400 \text{ K}$$

Since both  $\Delta H$  and  $\Delta S$  are positive, the reaction is spontaneous above 400 K.

## 2.8 pH, Acid Dissociation and Buffer Idea

For a weak acid:



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

For a buffer made of weak acid and its salt:

$$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

#### Solved Example: Buffer pH

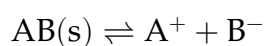
If  $\text{p}K_a = 4.75$ ,  $[\text{salt}] = 0.20 \text{ M}$  and  $[\text{acid}] = 0.10 \text{ M}$ :

$$\text{pH} = 4.75 + \log \frac{0.20}{0.10}$$

$$\text{pH} = 4.75 + \log 2 = 4.75 + 0.301 = 5.051$$

## 2.9 Solubility Product and Common Ion Effect

For a sparingly soluble salt AB:



$$K_{sp} = [A^+][B^-]$$

Precipitation occurs when ionic product exceeds  $K_{sp}$ .

| Condition                          | Meaning                             | Observation          |
|------------------------------------|-------------------------------------|----------------------|
| <b>Ionic product</b><br>$< K_{sp}$ | Unsaturated solution                | No precipitation     |
| <b>Ionic product</b><br>$= K_{sp}$ | Saturated solution                  | Equilibrium          |
| <b>Ionic product</b><br>$> K_{sp}$ | Supersaturated with respect to salt | Precipitation occurs |

## 2.10 Corrosion Prevention Methods

Corrosion can be reduced by blocking contact with environment or by making the metal cathodic.

| Method                     | Principle                               | Example                     |
|----------------------------|-----------------------------------------|-----------------------------|
| <b>Painting / coating</b>  | Barrier between metal and environment   | Paint, oil, polymer coating |
| <b>Galvanization</b>       | Zinc corrodes sacrificially before iron | Zn-coated iron sheets       |
| <b>Cathodic protection</b> | Metal to be protected is made cathode   | Pipelines, ship hulls       |
| <b>Alloying</b>            | Forms corrosion-resistant surface film  | Stainless steel             |
| <b>Inhibitors</b>          | Reduce anodic or cathodic reaction rate | Boiler-water additives      |

### Previous-Year Style Questions

1. Derive  $\Delta G^\circ = -RT \ln K$  and explain its significance.
2. Derive Nernst equation and solve a cell-potential numerical.
3. Explain water hardness, scale formation and boiler troubles.
4. Describe electrochemical corrosion of iron and prevention methods.
5. Explain Ellingham diagram and its use in metallurgy.

## Quick Revision — Module 4

Free Energy and Chemical Equilibria | BS-CH101

### Key Formulas at a Glance

| Formula / Rule                         | Meaning                                                |
|----------------------------------------|--------------------------------------------------------|
| $\Delta U = q + w$                     | First law of thermodynamics.                           |
| $\Delta G = \Delta H - T\Delta S$      | Gibbs free energy relation.                            |
| $\Delta G = \Delta G^\circ + RT \ln Q$ | Free energy at non-standard condition.                 |
| $\Delta G^\circ = -RT \ln K$           | Relation between free energy and equilibrium constant. |
| $\Delta G = -nFE$                      | Cell free energy and cell potential.                   |
| $E = E^\circ - \frac{RT}{nF} \ln Q$    | Nernst equation.                                       |
| $K_{sp} = [M^{m+}]^n [X^{n-}]^m$       | Solubility product expression.                         |

### One-Line Definitions

- **Entropy:** Measure of energy dispersal or randomness.
- **Gibbs free energy:** Energy available for useful work at constant temperature and pressure.
- **Equilibrium constant:** Ratio of product and reactant activities at equilibrium.
- **Nernst equation:** Relation between electrode potential and concentration.
- **Corrosion:** Electrochemical deterioration of a metal.
- **Ellingham diagram:** Plot of  $\Delta G^\circ$  for oxide formation versus temperature.

### Frequently Asked / Viva Questions

**Q1. State the first law of thermodynamics.**

Energy cannot be created or destroyed;  $\Delta U = q + w$ .

**Q2. What is the criterion of spontaneity using Gibbs free energy?**

$\Delta G < 0$  spontaneous,  $\Delta G = 0$  equilibrium,  $\Delta G > 0$  non-spontaneous.

**Q3. Write Gibbs free energy equation.**

$\Delta G = \Delta H - T\Delta S$ .

**Q4. Relate  $\Delta G^\circ$  and equilibrium constant.**

$\Delta G^\circ = -RT \ln K$ .

**Q5. How is cell emf related to free energy?**

$\Delta G = -nFE$ .

## Frequently Asked / Viva Questions (contd.)

**Q6. State the Nernst equation at 298 K.**

$$E = E^{\circ} - (0.0591/n) \log Q.$$

**Q7. What is solubility product?**

It is the product of ion concentrations in a saturated solution, each raised to stoichiometric power.

**Q8. Why does corrosion occur electrochemically?**

Metal oxidation and reduction of oxygen or hydrogen occur at different surface sites.

**Q9. What does an Ellingham diagram show?**

It shows temperature dependence of standard free energy of oxide formation.

**Q10. How is a reducing agent selected from Ellingham diagram?**

A reducing agent whose oxide line lies below the metal oxide line can reduce that oxide.

## Summary Reference Table

| Area                 | Key relation                      | Exam focus                         |
|----------------------|-----------------------------------|------------------------------------|
| Thermodynamics       | $\Delta G = \Delta H - T\Delta S$ | Spontaneity and equilibrium        |
| Chemical equilibrium | $\Delta G^{\circ} = -RT \ln K$    | Direction and extent of reaction   |
| Electrochemistry     | $\Delta G = -nFE$                 | Cell potential and Nernst equation |
| Metallurgy           | Ellingham diagram                 | Reduction feasibility              |

# Module 5

## Periodic Properties

---

### 1. Periodic Properties

---

**Periodic properties** are atomic and chemical properties that vary systematically with atomic number due to effective nuclear charge, shielding, and electronic configuration.

#### 1.1 Effective Nuclear Charge and Shielding

Outer electrons do not experience the full nuclear charge because inner electrons shield them. Effective nuclear charge is:

##### Effective Nuclear Charge

$$Z_{\text{eff}} = Z - \sigma$$

where  $Z$  is nuclear charge and  $\sigma$  is shielding constant.

#### 1.2 Penetration and Orbital Energy

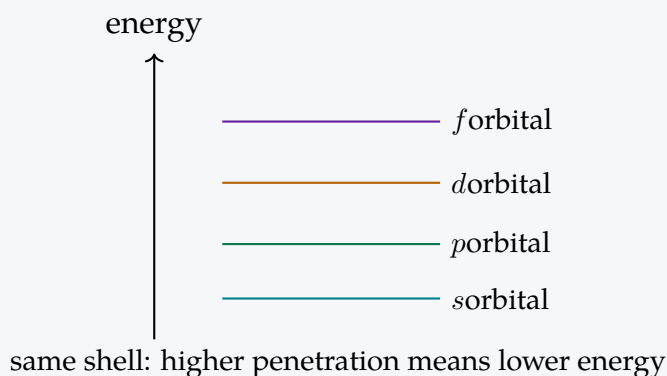
For the same principal shell, penetration order is:

$$s > p > d > f$$

More penetrating orbitals experience higher  $Z_{\text{eff}}$  and lower energy. Therefore in multi-electron atoms:

$$E(ns) < E(np) < E(nd) < E(nf)$$

## Penetration and Energy Order



## 1.3 Atomic Size, Ionization Energy, Electron Affinity and Electronegativity

Across a period,  $Z_{\text{eff}}$  increases, so atomic size decreases and ionization energy generally increases. Down a group, new shells are added, so atomic size increases and ionization energy decreases.

| Property           | Across period                   | Down group              |
|--------------------|---------------------------------|-------------------------|
| Atomic radius      | Decreases                       | Increases               |
| Ionization energy  | Generally increases             | Decreases               |
| Electron affinity  | Becomes more negative generally | Less negative generally |
| Electronegativity  | Increases                       | Decreases               |
| Metallic character | Decreases                       | Increases               |

## 1.4 Polarizability, Oxidation States and Coordination

Polarizability increases with size and diffuse electron cloud. Oxidation state depends on valence electron configuration. Coordination number and geometry depend on metal ion size, ligand size, and electron count.

| Coordination number | Common geometry              | Example type                    |
|---------------------|------------------------------|---------------------------------|
| 2                   | Linear                       | $[\text{Ag}(\text{NH}_3)_2]^+$  |
| 4                   | Tetrahedral or square planar | $[\text{Ni}(\text{CN})_4]^{2-}$ |
| 6                   | Octahedral                   | $[\text{Fe}(\text{CN})_6]^{3-}$ |

## 1.5 Hard and Soft Acids and Bases

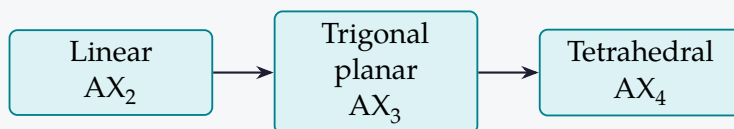
HSAB principle says hard acids prefer hard bases and soft acids prefer soft bases.

**Hard species** are small, highly charged, and weakly polarizable. **Soft species** are larger, lower charged, and highly polarizable.

## 1.6 Molecular Geometries

VSEPR theory predicts geometry from electron-pair repulsion around the central atom.

### Common VSEPR Shapes



### Exam Points

- Explain periodic trends using  $Z_{\text{eff}}$ , shielding, and shell number.
- For orbital energy, mention penetration order  $s > p > d > f$ .
- HSAB questions need definition plus examples.
- Geometry questions should mention electron pairs before shape.

## 2. Detailed Periodic Trend Analysis

### 2.1 Shielding Order and Slater-Type Reasoning

Shielding is not equal for all orbitals. Inner-shell electrons shield strongly, while electrons in the same shell shield only partially. Poor shielding by  $d$  and  $f$  electrons explains many irregular periodic trends.

| Electron type | Shielding ability                     | Consequence                          |
|---------------|---------------------------------------|--------------------------------------|
| $s$ electron  | Strong penetration and good shielding | Feels high $Z_{\text{eff}}$          |
| $p$ electron  | Moderate shielding                    | Higher energy than same-shell $s$    |
| $d$ electron  | Poor shielding                        | Causes transition-series contraction |
| $f$ electron  | Very poor shielding                   | Causes lanthanide contraction        |

### 2.2 Successive Ionization Energies

Successive ionization energies always increase because after removal of one electron, the remaining electrons are held more strongly by the positive ion.

$$\text{IE}_1 < \text{IE}_2 < \text{IE}_3 < \dots$$

A sudden jump in ionization energy indicates removal of an electron from an inner shell.

#### Solved Example: Identifying Valence Electrons

If successive ionization energies show the pattern:

$$580, \quad 1820, \quad 2750, \quad 11600 \text{ kJ mol}^{-1}$$

the large jump occurs after the third electron. Therefore the atom has **three valence electrons**.

### 2.3 Diagonal Relationship

Some second-period elements resemble third-period elements diagonally placed in the periodic table. Examples include Li–Mg, Be–Al, and B–Si. This occurs because size and charge density become comparable.

| Pair      | Common similarities                            | Reason                               |
|-----------|------------------------------------------------|--------------------------------------|
| Li and Mg | Form nitrides; carbonates decompose on heating | Similar charge density               |
| Be and Al | Amphoteric oxides and covalent halides         | Small size and high polarizing power |
| B and Si  | Covalent compounds and acidic oxides           | Similar electronegativity trend      |

### 2.4 Oxidation States and Inert Pair Effect

For heavier *p*-block elements, the outer  $ns^2$  electron pair becomes less available for bonding. This is called inert pair effect. It stabilizes lower oxidation states.

**Inert pair effect:** Reluctance of the outermost  $ns^2$  electron pair of heavier elements to participate in bonding.

| Group    | Higher oxidation state | Lower oxidation state stabilized down group |
|----------|------------------------|---------------------------------------------|
| Group 13 | +3                     | +1 in Tl                                    |
| Group 14 | +4                     | +2 in Pb                                    |
| Group 15 | +5                     | +3 in Bi                                    |

### 2.5 HSAB Applications

HSAB theory predicts stability of complexes and salts. Hard-hard and soft-soft combinations are more stable than hard-soft combinations.

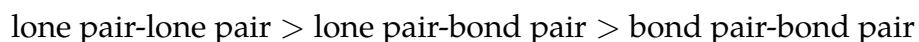
| Acid / base type | Examples                               | Preferred partner |
|------------------|----------------------------------------|-------------------|
| Hard acids       | $H^+$ , $Li^+$ , $Al^{3+}$ , $Fe^{3+}$ | Hard bases        |
| Soft acids       | $Ag^+$ , $Hg^{2+}$ , $Pt^{2+}$         | Soft bases        |
| Hard bases       | $F^-$ , $OH^-$ , $H_2O$ , $NH_3$       | Hard acids        |
| Soft bases       | $I^-$ , $S^{2-}$ , $CO$ , $PR_3$       | Soft acids        |

**Solved Example: HSAB Prediction**

$\text{Ag}^+$  is a soft acid. Between  $\text{F}^-$  and  $\text{I}^-$ , iodide is softer. Therefore  $\text{AgI}$  is more stable and less soluble than expected from simple charge arguments.

**2.6 VSEPR Shapes and Lone Pair Effect**

Lone pairs repel more strongly than bond pairs. The repulsion order is:



| Molecule type           | Shape              | Example                         |
|-------------------------|--------------------|---------------------------------|
| $\text{AX}_2$           | Linear             | $\text{BeCl}_2$ , $\text{CO}_2$ |
| $\text{AX}_3$           | Trigonal planar    | $\text{BF}_3$                   |
| $\text{AX}_3\text{E}$   | Trigonal pyramidal | $\text{NH}_3$                   |
| $\text{AX}_2\text{E}_2$ | Bent               | $\text{H}_2\text{O}$            |
| $\text{AX}_4$           | Tetrahedral        | $\text{CH}_4$                   |

**Periodic Properties: 10-Mark Answer Framework**

1. Define the property.
2. State the trend across period and down group.
3. Explain trend using  $Z_{\text{eff}}$ , shielding, and shell size.
4. Mention exceptions if relevant.
5. Add one example or application such as HSAB or VSEPR shape.

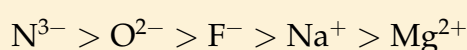
**2.7 Atomic and Ionic Radius: Detailed Trend**

Atomic radius decreases across a period because effective nuclear charge increases while electrons enter the same shell. Down a group, radius increases because a new shell is added.

| Species compared             | Trend                                           | Reason                                                     |
|------------------------------|-------------------------------------------------|------------------------------------------------------------|
| <b>Cations vs atoms</b>      | Cations are smaller                             | Electron loss reduces repulsion and may remove outer shell |
| <b>Anions vs atoms</b>       | Anions are larger                               | Extra electron increases electron-electron repulsion       |
| <b>Isoelectronic species</b> | Radius decreases with increasing nuclear charge | Same electrons pulled by larger $Z$                        |

**Solved Example: Isoelectronic Radius Order**

$\text{N}^{3-}$ ,  $\text{O}^{2-}$ ,  $\text{F}^-$ ,  $\text{Na}^+$  and  $\text{Mg}^{2+}$  are isoelectronic with 10 electrons. Higher nuclear charge pulls the same electron cloud more strongly:



## 2.8 Electron Affinity and Exceptions

Electron affinity generally becomes more negative across a period, but exceptions occur due to stable half-filled or filled configurations and small size.

| Exception                | Observation                            | Reason                                                 |
|--------------------------|----------------------------------------|--------------------------------------------------------|
| N less negative than C/O | N resists electron gain                | Stable half-filled $2p^3$ configuration                |
| F less negative than Cl  | Cl has more negative electron affinity | Very small F atom has high electron-electron repulsion |
| Noble gases              | Positive or very low tendency          | Stable closed shell                                    |

## 2.9 Electronegativity Scales

Electronegativity is not directly measured; it is assigned using scales. The Pauling scale is most common in basic chemistry.

| Scale         | Basis                                              | Use                                      |
|---------------|----------------------------------------------------|------------------------------------------|
| Pauling       | Bond energy differences                            | Most common qualitative scale            |
| Mulliken      | Average of ionization energy and electron affinity | Connects to atomic properties            |
| Allred-Rochow | Electrostatic attraction by nucleus                | Uses effective nuclear charge and radius |

## 2.10 Polarization and Fajan's Rules

Fajan's rules explain covalent character in ionic compounds. A small highly charged cation polarizes a large anion strongly, increasing covalent character.

### Fajan's Rules

- Smaller cation has greater polarizing power.
- Higher cation charge has greater polarizing power.
- Larger anion is more easily polarized.
- Cations with pseudo-noble gas configuration polarize more strongly.

### Solved Example: Covalent Character

$\text{AlCl}_3$  has more covalent character than  $\text{NaCl}$  because  $\text{Al}^{3+}$  is smaller and more highly charged than  $\text{Na}^+$ , so it polarizes  $\text{Cl}^-$  more strongly.

## 2.11 Coordination Number and Geometry in Complexes

Coordination number is the number of ligand donor atoms directly attached to the metal ion. Geometry depends on coordination number and electronic arrangement.

| Coordination number | Geometry      | Typical comment                                      |
|---------------------|---------------|------------------------------------------------------|
| 2                   | Linear        | Common for $\text{Ag}^+$ and $\text{Cu}^+$ complexes |
| 4                   | Tetrahedral   | Common when ligands are bulky or weak-field          |
| 4                   | Square planar | Common for $d^8$ metal ions like $\text{Pt}^{2+}$    |
| 6                   | Octahedral    | Most common geometry for transition-metal complexes  |

### Previous-Year Style Questions

1. Explain effective nuclear charge and shielding with periodic trends.
2. Discuss trends in atomic radius, ionization energy and electronegativity.
3. Explain penetration of  $s$ ,  $p$ ,  $d$ ,  $f$  orbitals and its effect on energy.
4. State HSAB principle and predict stable acid-base combinations.
5. Explain VSEPR shapes and coordination geometries with examples.

## 2.12 Common Exceptions and Exam Traps

Periodic trends are general rules, but exam questions often test exceptions. Mentioning the reason behind exceptions improves answer quality.

| Trend exception                   | Observation                                  | Reason                                                    |
|-----------------------------------|----------------------------------------------|-----------------------------------------------------------|
| <b>Be and B ionization energy</b> | $\text{IE}(\text{Be}) > \text{IE}(\text{B})$ | Electron removed from B enters higher-energy $2p$ orbital |
| <b>N and O ionization energy</b>  | $\text{IE}(\text{N}) > \text{IE}(\text{O})$  | N has stable half-filled $2p^3$ arrangement               |
| <b>Ga smaller than Al anomaly</b> | Size does not increase as expected           | Poor shielding by filled $3d$ electrons                   |
| <b>Lanthanide contraction</b>     | Similar radii of $4d$ and $5d$ elements      | Poor shielding by $4f$ electrons                          |

### Exam Answer: Why $\text{IE}(\text{N}) > \text{IE}(\text{O})$ ?

Nitrogen has electronic configuration  $1s^2 2s^2 2p^3$ , where the three  $p$  electrons are half-filled and relatively stable. Oxygen has  $1s^2 2s^2 2p^4$ , so one  $p$  orbital contains paired electrons. Removing one electron from oxygen reduces electron-electron repulsion, so oxygen has lower first ionization energy than nitrogen.

## Quick Revision — Module 5

Periodic Properties | BS-CH101

### Key Formulas at a Glance

| Formula / Rule                  | Meaning                                                                |
|---------------------------------|------------------------------------------------------------------------|
| $Z_{\text{eff}} = Z - \sigma$   | Effective nuclear charge felt by valence electron.                     |
| $s > p > d > f$                 | Penetration order within a shell.                                      |
| $E(ns) < E(np) < E(nd) < E(nf)$ | Orbital energy order for same principal shell in multi-electron atoms. |
| $IE_1 < IE_2 < IE_3$            | Successive ionization energies increase.                               |
| $\mu = \alpha E$                | Induced dipole moment relation; $\alpha$ is polarizability.            |

### One-Line Definitions

- **Effective nuclear charge:** Net positive charge experienced by an electron.
- **Shielding:** Reduction of nuclear attraction due to inner electrons.
- **Penetration:** Ability of an orbital to approach the nucleus.
- **Ionization energy:** Energy required to remove an electron from gaseous atom.
- **Electron affinity:** Energy change when an electron is added to gaseous atom.
- **Electronegativity:** Tendency of atom to attract bonded electron pair.
- **HSAB principle:** Hard acids bind hard bases and soft acids bind soft bases.

### Frequently Asked / Viva Questions

**Q1. Define effective nuclear charge.**

It is the net nuclear charge felt by an electron after shielding.

**Q2. Why does atomic radius decrease across a period?**

$Z_{\text{eff}}$  increases and pulls electrons closer.

**Q3. Why does ionization energy decrease down a group?**

Atomic size and shielding increase, so valence electrons are removed more easily.

**Q4. Write the penetration order.**

$s > p > d > f$ .

**Q5. Why is an  $s$  orbital lower in energy than a  $p$  orbital of same shell?**

The  $s$  orbital penetrates closer to the nucleus and feels greater  $Z_{\text{eff}}$ .

## Frequently Asked / Viva Questions (contd.)

**Q6. What is polarizability?**

Ease with which an electron cloud is distorted by an electric field.

**Q7. What is HSAB principle?**

Hard acids prefer hard bases and soft acids prefer soft bases.

**Q8. Give examples of hard and soft bases.**

Hard bases:  $F^-$ ,  $OH^-$ ; soft bases:  $I^-$ ,  $S^{2-}$ .

**Q9. What decides coordination geometry?**

Coordination number, ligand field, metal ion size, and electron configuration.

**Q10. What is VSEPR theory used for?**

It predicts molecular geometry from repulsion among electron pairs.

## Summary Reference Table

| Trend         | Reason                            | Result                          |
|---------------|-----------------------------------|---------------------------------|
| Across period | $Z_{\text{eff}}$ increases        | Size decreases, IE increases    |
| Down group    | Shell number increases            | Size increases, IE decreases    |
| Penetration   | $s$ enters closest to nucleus     | $s$ orbital has lower energy    |
| HSAB          | Charge density and polarizability | Predicts stable acid-base pairs |

# Module 6

## Stereochemistry

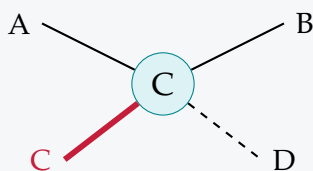
### 1. Stereochemistry

**Stereochemistry** is the study of three-dimensional arrangement of atoms in molecules and how this arrangement affects physical, chemical, and biological properties.

#### 1.1 Representations of Three-Dimensional Structures

Three-dimensional structures are represented using wedge-dash, Fischer projection, Newman projection, and sawhorse projection.

##### Wedge-Dash Representation Around a Tetrahedral Carbon



solid wedge: coming out; dashed bond: going back

#### 1.2 Structural Isomers and Stereoisomers

Structural isomers differ in connectivity, whereas stereoisomers have same connectivity but different spatial arrangement.

| Type                  | Meaning                                       | Example                         |
|-----------------------|-----------------------------------------------|---------------------------------|
| Chain isomerism       | Different carbon skeleton                     | n-butane and isobutane          |
| Position isomerism    | Functional group at different position        | 1-propanol and 2-propanol       |
| Geometrical isomerism | Different arrangement around double bond/ring | cis-2-butene and trans-2-butene |
| Optical isomerism     | Non-superimposable mirror images              | lactic acid enantiomers         |

### 1.3 Chirality, Enantiomers and Diastereomers

A molecule is chiral if it is not superimposable on its mirror image. A common cause is a carbon attached to four different groups.

**Enantiomers** are non-superimposable mirror-image stereoisomers. **Diastereomers** are stereoisomers that are not mirror images.

### 1.4 Optical Activity and Absolute Configuration

Optically active substances rotate plane-polarized light. Absolute configuration is assigned by Cahn-Ingold-Prelog rules.

#### Specific Rotation

$$[\alpha]_{\lambda}^T = \frac{\alpha}{lc}$$

where  $\alpha$  is observed rotation in degrees,  $l$  is path length in dm, and  $c$  is concentration in  $\text{g mL}^{-1}$ .

### 1.5 Elements of Symmetry

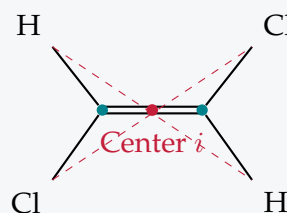
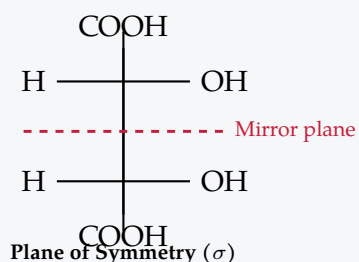
To understand chirality and optical activity, we must analyze the symmetry elements of a molecule. A molecule is chiral (and thus optically active) if and only if it lacks an **alternating axis of symmetry** ( $S_n$ ). In practice, for most organic molecules, this is equivalent to lacking a **plane of symmetry** ( $\sigma$ ) and a **center of symmetry** ( $i$ ).

#### Symmetry Elements

The three primary elements of symmetry are:

- **Plane of Symmetry** ( $\sigma$ ): An imaginary plane that bisects a molecule into two halves that are mirror images of each other. Examples include meso-tartaric acid and chlorobenzene.
- **Center of Symmetry** ( $i$ ): A point within a molecule such that any line drawn from any atom through this point meets an identical atom at an equal distance on the opposite side. Examples include trans-1,2-dichloroethylene and trans-1,4-dimethylcyclohexane.
- **Alternating Axis of Symmetry** ( $S_n$ ): An axis such that rotation of the molecule by  $360^\circ/n$  followed by reflection across a plane perpendicular to the axis results in a configuration identical to the original.

## Symmetry Elements in Common Molecules



## Chirality and Symmetry Rule

A molecule is chiral and optically active if and only if it is **dissymmetric** (lacks an alternating axis of symmetry,  $S_n$ ).

$$\text{Chirality} \iff \text{Absence of } S_n \quad (\text{Practically: absence of } \sigma \text{ and } i)$$

Note: A molecule can have a simple axis of symmetry ( $C_n$ ) and still be chiral. For example, trans-1,2-dimethylcyclopropane has a  $C_2$  axis but no  $\sigma$  or  $i$ , making it chiral and optically active.

## 1.6 Conformational Analysis

Conformations arise by rotation about single bonds. Ethane has staggered and eclipsed conformations; staggered is more stable due to lower torsional strain.

## Ethane Conformations: Staggered and Eclipsed

Staggered



Eclipsed



## 1.7 Isomerism in Transition Metal Compounds

Coordination compounds show geometrical, optical, linkage, ionization, and coordination isomerism. Square planar and octahedral complexes commonly show cis-trans isomerism.

## Exam Points

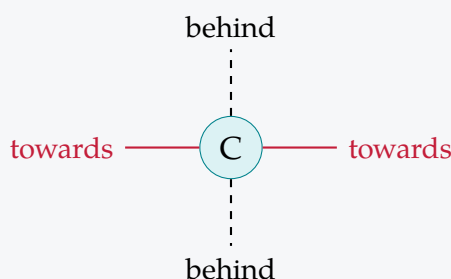
- Always distinguish structural isomerism from stereoisomerism.
- In optical isomerism, mention chirality and non-superimposable mirror images.
- CIP assignment needs priority order before clockwise/anticlockwise decision.
- For conformational analysis, compare stability using torsional strain.

## 2. Detailed Stereochemical Analysis

### 2.1 Fischer Projection Rules

Fischer projections are used mainly for molecules with chiral carbon chains. Horizontal bonds come out of the plane toward the viewer; vertical bonds go behind the plane.

#### Fischer Projection Convention



#### Rules for Fischer Projections

- Rotation by  $180^\circ$  gives the same molecule.
- Rotation by  $90^\circ$  gives a different configuration and is not allowed.
- Interchanging any two groups once inverts configuration.
- Interchanging two pairs of groups gives the original configuration.

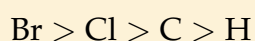
### 2.2 Assigning R and S Configuration

Cahn-Ingold-Prelog priority rules:

1. Atom directly attached with higher atomic number gets higher priority.
2. If first atoms are same, compare the next set of attached atoms.
3. Multiple bonds are treated as equivalent single bonds to duplicate atoms.
4. View the molecule with the lowest priority group pointing away.
5. Clockwise order  $1 \rightarrow 2 \rightarrow 3$  is *R*; anticlockwise is *S*.

#### Worked Method: R/S Assignment

For a chiral carbon attached to Br, Cl,  $\text{CH}_3$ , and H:



Place H away from the viewer. If the order  $\text{Br} \rightarrow \text{Cl} \rightarrow \text{CH}_3$  is clockwise, the configuration is *R*; if anticlockwise, it is *S*.

### 2.3 Geometrical Isomerism: E/Z and Cis/Trans

Geometrical isomerism occurs due to restricted rotation around a double bond or within rings. Cis/trans notation is used for simple cases; E/Z notation is used when each double-bond carbon has two different groups.

| Notation     | Meaning                                  | Use case                    |
|--------------|------------------------------------------|-----------------------------|
| <b>Cis</b>   | Similar groups on same side              | Simple alkenes or rings     |
| <b>Trans</b> | Similar groups on opposite sides         | Simple alkenes or rings     |
| <b>E</b>     | Higher-priority groups on opposite sides | General alkenes by CIP rule |
| <b>Z</b>     | Higher-priority groups on same side      | General alkenes by CIP rule |

### 2.4 Meso Compounds

A meso compound contains chiral centres but is optically inactive due to internal symmetry. It is superimposable on its mirror image because optical rotations of two halves cancel internally.

**Meso compound:** An achiral molecule with two or more stereocentres and an internal plane of symmetry.

### 2.5 Conformational Analysis of Butane

Butane has anti, gauche, and eclipsed conformations. Anti is most stable because bulky methyl groups are far apart.

| Conformation            | Dihedral angle between methyl groups | Relative stability                   |
|-------------------------|--------------------------------------|--------------------------------------|
| <b>Anti staggered</b>   | 180°                                 | Most stable                          |
| <b>Gauche staggered</b> | 60°                                  | Less stable due to steric repulsion  |
| <b>Eclipsed</b>         | 0° or 120°                           | Least stable due to torsional strain |

#### Specific Rotation Numerical

If observed rotation  $\alpha = +6^\circ$ , path length  $l = 2 \text{ dm}$ , and concentration  $c = 0.5 \text{ g mL}^{-1}$ :

$$[\alpha] = \frac{\alpha}{lc} = \frac{6}{2 \times 0.5} = +6^\circ$$

### 2.6 Coordination Compound Stereoisomerism

Square planar complexes of type  $\text{MA}_2\text{B}_2$  show cis-trans isomerism. Octahedral complexes may show cis-trans, fac-mer, and optical isomerism depending on ligand arrangement.

| Complex type                      | Possible isomerism | Example pattern                            |
|-----------------------------------|--------------------|--------------------------------------------|
| <b>Square planar</b><br>$MA_2B_2$ | Cis-trans          | $[Pt(NH_3)_2Cl_2]$                         |
| <b>Octahedral</b><br>$MA_4B_2$    | Cis-trans          | Two B ligands adjacent or opposite         |
| <b>Octahedral</b><br>$MA_3B_3$    | fac-mer            | Three same ligands on one face or meridian |

### Stereochemistry Answer Checklist

1. Identify whether connectivity changes or only spatial arrangement changes.
2. Check for chirality, plane of symmetry, and mirror-image relation.
3. Use CIP rules for R/S or E/Z configuration.
4. Mention optical activity only after checking symmetry.

## 2.7 Number of Stereoisomers

For a molecule with  $n$  independent chiral centres, the maximum number of stereoisomers is:

$$2^n$$

If the molecule has internal symmetry, the actual number is less due to meso forms.

### Solved Example: Number of Stereoisomers

A molecule has two chiral centres and no plane of symmetry.

$$\text{Maximum stereoisomers} = 2^2 = 4$$

These are arranged as two pairs of enantiomers. If a plane of symmetry exists, one or more meso forms reduce the total count.

## 2.8 Resolution of Racemic Mixtures

A racemic mixture contains equal amounts of two enantiomers and is optically inactive. Resolution means separation of enantiomers.

| Method                            | Principle                                     | Use                                       |
|-----------------------------------|-----------------------------------------------|-------------------------------------------|
| <b>Mechanical separation</b>      | Physical separation of crystals               | Rare; possible if crystals differ visibly |
| <b>Chemical resolution</b>        | Formation of diastereomeric salts             | Common laboratory method                  |
| <b>Biochemical resolution</b>     | Enzyme reacts with one enantiomer faster      | High selectivity                          |
| <b>Chromatographic resolution</b> | Chiral stationary phase separates enantiomers | Analytical and preparative use            |

## 2.9 Optical Purity and Enantiomeric Excess

Optical purity compares observed rotation with rotation of the pure enantiomer.

### Optical Purity

$$\text{Optical purity} = \frac{\alpha_{\text{observed}}}{\alpha_{\text{pure}}} \times 100\%$$

For a two-enantiomer mixture, optical purity is equal to enantiomeric excess.

### Solved Example: Enantiomeric Excess

If pure enantiomer has specific rotation  $+40^\circ$  and mixture shows  $+20^\circ$ :

$$\text{optical purity} = \frac{20}{40} \times 100 = 50\%$$

So enantiomeric excess is 50%. The mixture contains 75% major enantiomer and 25% minor enantiomer.

## 2.10 Sequence Rules for Double Bonds: E/Z Worked Method

For each double-bond carbon, assign priority to its two substituents using CIP rules. If the two higher-priority groups are on the same side, the alkene is *Z*; if opposite, it is *E*.

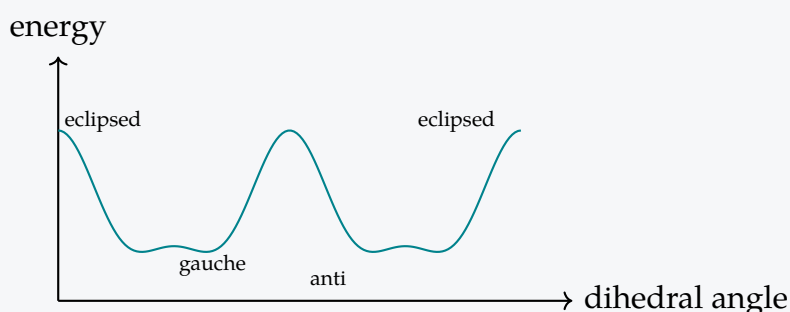
### Worked Method: E/Z Assignment

For  $\text{CH}_3\text{CH}=\text{CHCl}$ , compare substituents on each double-bond carbon. On the left carbon,  $\text{CH}_3$  has higher priority than H. On the right carbon, Cl has higher priority than H. If  $\text{CH}_3$  and Cl are on opposite sides, the configuration is *E*; if same side, it is *Z*.

## 2.11 Energy Profile of Butane Rotation

Rotation about the central C–C bond of butane gives different conformations. Anti is most stable; fully eclipsed is least stable.

### Qualitative Energy Profile for Butane Rotation



## 2.12 Stereochemistry in Biological and Drug Action

Many drug molecules are chiral, and different enantiomers can show different biological activity because enzymes and receptors are chiral environments.

| Point                  | Importance                                                       |
|------------------------|------------------------------------------------------------------|
| Different receptor fit | One enantiomer may bind better than the other                    |
| Different metabolism   | Enzymes may break down enantiomers at different rates            |
| Different toxicity     | One enantiomer may be useful while the other causes side effects |
| Quality control        | Optical purity matters in pharmaceutical synthesis               |

#### Previous-Year Style Questions

1. Explain chirality, enantiomers and diastereomers with examples.
2. State CIP rules and assign R/S configuration for a given molecule.
3. Explain optical activity, specific rotation and racemic mixture.
4. Compare conformations of ethane and butane.
5. Discuss geometrical and optical isomerism in coordination compounds.

## Quick Revision — Module 6

Stereochemistry | BS-CH101

### Key Formulas at a Glance

| Formula / Rule                                                    | Meaning                                              |
|-------------------------------------------------------------------|------------------------------------------------------|
| $[\alpha]_{\lambda}^T = \frac{\alpha}{lc}$                        | Specific rotation of optically active compound.      |
| <b>enantiomeric excess</b> = $\frac{ R - S }{R + S} \times 100\%$ | Optical purity of a mixture.                         |
| $\Delta E = E_{\text{eclipsed}} - E_{\text{staggered}}$           | Torsional energy barrier in conformational analysis. |
| <b>CIP priority</b>                                               | Higher atomic number gets higher priority.           |

### One-Line Definitions

- **Stereoisomers:** Same connectivity but different spatial arrangement.
- **Chiral molecule:** Molecule not superimposable on its mirror image.
- **Enantiomers:** Non-superimposable mirror-image isomers.
- **Diastereomers:** Stereoisomers that are not mirror images.
- **Optical activity:** Ability to rotate plane-polarized light.
- **Conformation:** Spatial arrangement formed by rotation about single bond.
- **Racemic mixture:** Equimolar mixture of two enantiomers.

### Frequently Asked / Viva Questions

**Q1. What is stereochemistry?**

Study of three-dimensional arrangement of atoms in molecules.

**Q2. What is chirality?**

Non-superimposability of a molecule on its mirror image.

**Q3. Define enantiomers.**

Enantiomers are non-superimposable mirror-image stereoisomers.

**Q4. Define diastereomers.**

Diastereomers are stereoisomers that are not mirror images.

**Q5. What is optical activity?**

Rotation of plane-polarized light by a chiral substance.

## Frequently Asked / Viva Questions (contd.)

**Q6. What is a racemic mixture?**

A 1:1 mixture of enantiomers having zero net optical rotation.

**Q7. What does R/S configuration indicate?**

It gives absolute three-dimensional arrangement around a chiral centre.

**Q8. Why is staggered ethane more stable than eclipsed ethane?**

Staggered conformation has minimum torsional repulsion.

**Q9. Name isomerism in coordination compounds.**

Geometrical, optical, linkage, ionization, and coordination isomerism.

**Q10. What is geometrical isomerism?**

Isomerism due to restricted rotation causing cis-trans or fac-mer forms.

**Q11. What are the elements of symmetry that determine chirality?**

A plane of symmetry ( $\sigma$ ), center of symmetry ( $i$ ), and alternating axis of symmetry ( $S_n$ ). A molecule is chiral if and only if it lacks any  $S_n$  axis.

**Q12. Can a molecule containing chiral carbon atoms be optically inactive?**

Yes, meso compounds contain chiral carbons but are optically inactive because they have an internal plane of symmetry ( $\sigma$ ), which causes internal compensation of optical rotation.

## Summary Reference Table

| Pair                      | Relation                   | Key difference                      |
|---------------------------|----------------------------|-------------------------------------|
| <b>Structural isomers</b> | Different connectivity     | Different bonding sequence          |
| <b>Enantiomers</b>        | Mirror images              | Equal and opposite optical rotation |
| <b>Diastereomers</b>      | Not mirror images          | Different physical properties       |
| <b>Conformers</b>         | Rotation about single bond | Interconvert without bond breaking  |

# Module 7

## Organic Reactions and Drug Molecule Synthesis

### 1. Organic Reactions and Drug Molecule Synthesis

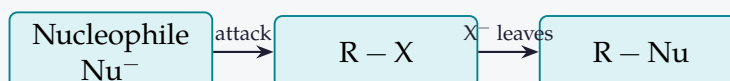
**Organic reaction mechanism** explains the step-by-step breaking and making of bonds during conversion of reactants into products. It identifies intermediates, reagents, and electron movement.

#### 1.1 Substitution Reactions

In substitution, one atom or group is replaced by another. Nucleophilic substitution is common in alkyl halides.

| Type             | Main feature                       | Favoured by                                  |
|------------------|------------------------------------|----------------------------------------------|
| S <sub>N</sub> 1 | Two-step, carbocation intermediate | Tertiary substrate, polar protic solvent     |
| S <sub>N</sub> 2 | One-step backside attack           | Methyl/primary substrate, strong nucleophile |

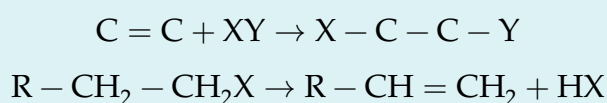
#### SN2 Backside Attack Concept



#### 1.2 Addition and Elimination Reactions

Addition reactions add atoms across multiple bonds. Elimination reactions remove atoms from adjacent carbons to form a double bond.

#### Typical Patterns



### 1.3 Oxidation, Reduction, Cyclization and Ring Opening

Oxidation increases C-O, C-N, or C-X bonds or decreases C-H bonds. Reduction increases C-H bonds or removes electronegative atoms. Cyclization forms a ring; ring opening converts cyclic compounds into open-chain products.

| Reaction     | Meaning                            | Example idea             |
|--------------|------------------------------------|--------------------------|
| Oxidation    | Gain of oxygen or loss of hydrogen | Alcohol to aldehyde/acid |
| Reduction    | Gain of hydrogen or loss of oxygen | Carbonyl to alcohol      |
| Cyclization  | Ring formation                     | Intramolecular attack    |
| Ring opening | Cleavage of a ring bond            | Epoxide hydrolysis       |

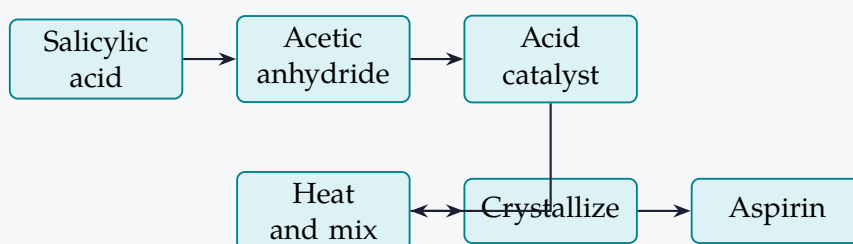
### 1.4 Synthesis of a Common Drug Molecule: Aspirin

Aspirin (acetylsalicylic acid) is prepared by acetylation of salicylic acid using acetic anhydride in acidic medium.

#### Aspirin Synthesis Reaction



#### Flow Diagram for Aspirin Preparation



### 1.5 Synthesis of a Common Drug Molecule: Paracetamol

Paracetamol (acetaminophen or *p*-acetamidophenol) is a widely used analgesic (pain reliever) and antipyretic (fever reducer). It is synthesized from phenol via a three-step pathway.

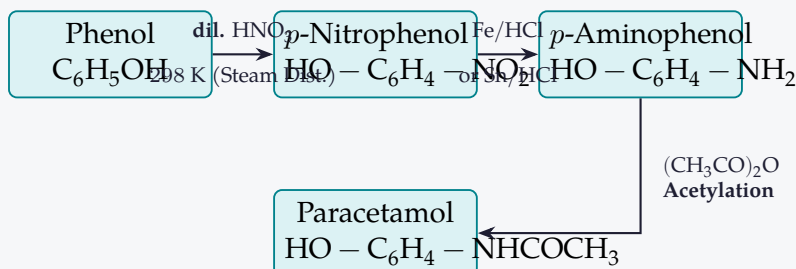
#### Paracetamol Synthesis Reactions

The three steps involved in the industrial/laboratory synthesis are:

- Nitration of Phenol:** Phenol reacts with dilute nitric acid ( $\text{HNO}_3$ ) at low temperature ( $\sim 298 \text{ K}$ ) to yield a mixture of *o*-nitrophenol and *p*-nitrophenol. The para isomer is separated by steam distillation (since the ortho isomer is steam-volatile due to intramolecular hydrogen bonding).
- Reduction to *p*-Aminophenol:** *p*-nitrophenol is reduced to *p*-aminophenol using reducing agents like iron and hydrochloric acid ( $\text{Fe}/\text{HCl}$ ) or tin and hydrochloric acid ( $\text{Sn}/\text{HCl}$ ).
- Acetylation:** *p*-aminophenol is reacted with acetic anhydride ( $(\text{CH}_3\text{CO})_2\text{O}$ ) in an

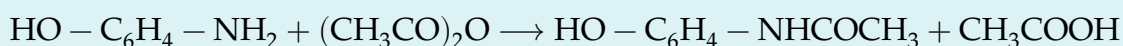
aqueous or weakly acidic/basic medium. The amino group ( $-\text{NH}_2$ ) undergoes selective acetylation to yield Paracetamol.

#### Reaction Scheme: Phenol to Paracetamol



#### Selective Acetylation Mechanism (Chemoselectivity)

In *p*-aminophenol, both the amino group ( $-\text{NH}_2$ ) and the hydroxyl group ( $-\text{OH}$ ) are nucleophilic. However, nitrogen is less electronegative than oxygen, making the lone pair on nitrogen much more nucleophilic and available for attack. Therefore, acetylation occurs exclusively at the nitrogen atom:



#### Mechanism in Words:

1. The nucleophilic nitrogen lone pair of *p*-aminophenol attacks one of the carbonyl carbons of acetic anhydride.
2. A tetrahedral intermediate is formed.
3. The carbonyl double bond is restored, and an acetate ion departs as the leaving group.
4. Deprotonation of the nitrogen atom gives Paracetamol (*p*-acetamidophenol) and acetic acid.

### 1.6 Mechanistic Exam Language

For mechanism-based answers, identify electrophile, nucleophile, leaving group, intermediate, and product. Use curved-arrow reasoning in words even when detailed curved arrows are not drawn.

#### Exam Points

- Compare  $\text{S}_{\text{N}}1$  and  $\text{S}_{\text{N}}2$  using step, intermediate, rate law, and stereochemistry.
- For addition and elimination, write the general reaction pattern.
- For oxidation/reduction, state change in C-H and C-O bonds.
- For aspirin synthesis, mention salicylic acid, acetic anhydride, acid catalyst, and acetylation.

## 2. Detailed Organic Reaction Notes and Synthesis Planning

### 2.1 Reaction Intermediates

Organic reactions often proceed through reactive intermediates. Stability of intermediates decides product distribution and rate.

| Intermediate | Nature                                         | Stability trend                                  |
|--------------|------------------------------------------------|--------------------------------------------------|
| Carbocation  | Electron-deficient carbon with positive charge | $3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$    |
| Carbanion    | Electron-rich carbon with negative charge      | Stabilized by electron-withdrawing groups        |
| Free radical | Species with unpaired electron                 | $3^\circ > 2^\circ > 1^\circ > \text{CH}_3\cdot$ |
| Carbene      | Neutral divalent carbon species                | Highly reactive; addition to alkenes             |

### 2.2 Detailed Comparison of $\text{S}_\text{N}1$ and $\text{S}_\text{N}2$

| Point           | $\text{S}_\text{N}1$          | $\text{S}_\text{N}2$         |
|-----------------|-------------------------------|------------------------------|
| Mechanism       | Two-step                      | One-step concerted           |
| Rate law        | $k[\text{RX}]$                | $k[\text{RX}][\text{Nu}^-]$  |
| Intermediate    | Carbocation                   | No intermediate              |
| Substrate       | $3^\circ > 2^\circ > 1^\circ$ | methyl $> 1^\circ > 2^\circ$ |
| Solvent         | Polar protic                  | Polar aprotic                |
| Stereochemistry | Racemization possible         | Inversion of configuration   |

#### Solved Concept: Choosing $\text{S}_\text{N}1$ or $\text{S}_\text{N}2$

Tertiary butyl chloride reacts with water mainly by  $\text{S}_\text{N}1$  because a stable tertiary carbocation can form and water is a weak nucleophile in polar protic medium. Methyl bromide reacts with  $\text{OH}^-$  mainly by  $\text{S}_\text{N}2$  because backside attack is unhindered.

### 2.3 Electrophilic Addition to Alkenes

Alkenes undergo electrophilic addition because the pi bond is electron rich. Addition of HX follows Markovnikov rule in normal ionic addition.

**Markovnikov rule:** In addition of HX to an unsymmetrical alkene, hydrogen attaches to the carbon already having more hydrogen atoms, and X attaches to the more substituted carbon.

**Addition of HBr to Propene**

The major product is 2-bromopropane due to formation of the more stable secondary carbocation.

**2.4 Elimination: E1 and E2**

Elimination reactions form alkenes. E1 proceeds through carbocation; E2 occurs in one step with base removing beta-hydrogen as leaving group departs.

| Point               | E1                         | E2                                            |
|---------------------|----------------------------|-----------------------------------------------|
| Steps               | Two-step                   | One-step                                      |
| Intermediate        | Carbocation                | None                                          |
| Base strength       | Weak base possible         | Strong base required                          |
| Substrate           | Tertiary favoured          | Primary, secondary, tertiary possible         |
| Stereochemical need | No strict anti requirement | Anti-periplanar H and leaving group preferred |

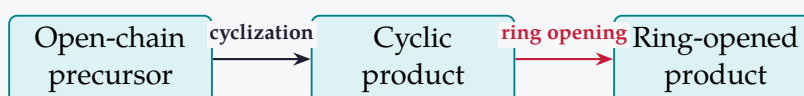
**2.5 Oxidation and Reduction Reagents**

Remembering reagent function is more useful than memorizing long mechanisms for basic engineering chemistry.

| Reagent                                                       | Type                   | Common use                                           |
|---------------------------------------------------------------|------------------------|------------------------------------------------------|
| KMnO <sub>4</sub>                                             | Strong oxidizing agent | Alkene oxidation, alcohol oxidation                  |
| K <sub>2</sub> Cr <sub>2</sub> O <sub>7</sub> /H <sup>+</sup> | Oxidizing agent        | Primary alcohol to acid, secondary alcohol to ketone |
| LiAlH <sub>4</sub>                                            | Strong reducing agent  | Carbonyl and acid derivatives to alcohols            |
| NaBH <sub>4</sub>                                             | Mild reducing agent    | Aldehyde/ketone to alcohol                           |
| H <sub>2</sub> /Ni                                            | Catalytic reduction    | Alkene to alkane                                     |

**2.6 Cyclization and Ring Opening**

Cyclization is favoured when the reacting ends of a molecule can form a stable five- or six-membered ring. Ring opening is important for strained rings like epoxides.

**Cyclization and Ring Opening Concept**

## 2.7 Aspirin Synthesis: Mechanism and Purification

In aspirin preparation, salicylic acid provides the phenolic —OH group and acetic anhydride provides the acetyl group. Acid catalyst increases electrophilicity of acetic anhydride. The product is purified by recrystallization.

1. Protonation of acetic anhydride increases electrophilic character of carbonyl carbon.
2. Phenolic oxygen of salicylic acid attacks the carbonyl carbon.
3. Acetate group leaves and deprotonation gives aspirin.
4. Cold water helps crystallize aspirin because aspirin is less soluble in cold water.

### Aspirin Yield Calculation

If theoretical yield is 5.00 g and actual yield is 3.80 g:

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

$$\% \text{ yield} = \frac{3.80}{5.00} \times 100 = 76\%$$

### Organic Reaction Answer Checklist

1. Identify substrate, reagent, nucleophile/electrophile, and leaving group.
2. Classify the reaction as substitution, addition, elimination, oxidation, reduction, cyclization, or ring opening.
3. Write rate law or regioselectivity rule where applicable.
4. Mention major product and reason for its formation.
5. For drug synthesis, include reactants, catalyst, reaction type, purification, and yield idea.

## 2.8 Reaction Map for Basic Organic Transformations

Most exam questions ask how one functional group changes into another. The following map gives the common first-year transformations.

| Conversion              | Typical reagent                                                                    | Reaction class            |
|-------------------------|------------------------------------------------------------------------------------|---------------------------|
| Alkene to alkane        | H <sub>2</sub> /Ni                                                                 | Reduction / hydrogenation |
| Alkene to alcohol       | H <sub>2</sub> O/H <sup>+</sup>                                                    | Addition                  |
| Alcohol to alkene       | conc. H <sub>2</sub> SO <sub>4</sub> , heat                                        | Elimination / dehydration |
| Primary alcohol to acid | KMnO <sub>4</sub> or K <sub>2</sub> Cr <sub>2</sub> O <sub>7</sub> /H <sup>+</sup> | Oxidation                 |
| Aldehyde to alcohol     | NaBH <sub>4</sub>                                                                  | Reduction                 |
| Alkyl halide to alcohol | aqueous KOH                                                                        | Nucleophilic substitution |

## 2.9 Curly-Arrow Logic in Words

Even when curved arrows are not drawn, the answer should describe electron flow clearly:

1. A nucleophile donates an electron pair.
2. An electrophile accepts an electron pair.
3. A leaving group departs with the bonding electron pair.
4. Formation of a stable intermediate or transition state controls the major product.

### Worked Mechanism in Words: Alcohol to Alkene

In acid-catalysed dehydration of an alcohol, the  $\text{-OH}$  group is first protonated to form a better leaving group, water. Water leaves to form a carbocation. A base removes a beta-hydrogen, and a double bond forms. The major alkene is usually the more substituted alkene according to Zaitsev rule.

## 2.10 Zaitsev and Hofmann Elimination

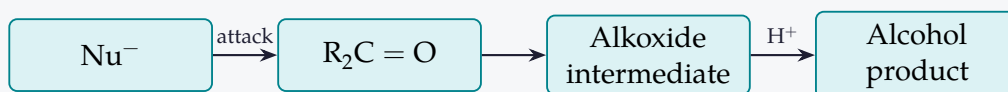
When more than one alkene can form, regioselectivity matters.

| Rule         | Major product           | Typical condition                             |
|--------------|-------------------------|-----------------------------------------------|
| Zaitsev rule | More substituted alkene | Small strong base or normal elimination       |
| Hofmann rule | Less substituted alkene | Bulky base or quaternary ammonium elimination |

## 2.11 Addition to Carbonyl Compounds

The carbonyl group is polar. Carbon is electrophilic and oxygen is nucleophilic/basic. Nucleophiles attack the carbonyl carbon to form addition products.

### Carbonyl Addition Concept



## 2.12 Green Chemistry Note for Drug Synthesis

Engineering chemistry answers may include practical synthesis quality. A good drug synthesis should be high-yielding, selective, safe, and easy to purify.

| Green chemistry point | Meaning in aspirin synthesis                                |
|-----------------------|-------------------------------------------------------------|
| Atom economy          | Prefer maximum incorporation of reactant atoms into product |
| Safer solvent         | Use minimum harmful solvent during purification             |
| Energy efficiency     | Avoid unnecessary heating time                              |
| Waste reduction       | Reduce excess reagent and by-product handling               |
| Product purity        | Recrystallization improves purity for medicinal use         |

### 2.13 Tests for Functional Groups

Short practical-style questions may ask how a functional group is detected.

| Functional group | Test                 | Positive observation                   |
|------------------|----------------------|----------------------------------------|
| Alkene           | Bromine water        | Decolorization                         |
| Aldehyde         | Tollens' reagent     | Silver mirror                          |
| Carboxylic acid  | Sodium bicarbonate   | Brisk effervescence of CO <sub>2</sub> |
| Phenol           | Ferric chloride test | Violet or coloured complex             |
| Alcohol          | Esterification       | Fruity smell of ester                  |

#### Previous-Year Style Questions

1. Compare S<sub>N</sub>1 and S<sub>N</sub>2 mechanisms with stereochemical outcome.
2. Explain electrophilic addition to propene using Markovnikov rule.
3. Compare E1 and E2 elimination reactions.
4. Write short notes on oxidation and reduction reagents in organic chemistry.
5. Describe aspirin synthesis, purification and percentage yield calculation.

## Quick Revision — Module 7

Organic Reactions and Drug Molecule Synthesis | BS-CH101

### Key Formulas at a Glance

| Formula / Rule                                   | Meaning                                                                                                                                            |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{rate}_{S_N1} = k[\text{RX}]$              | First-order nucleophilic substitution.                                                                                                             |
| $\text{rate}_{S_N2} = k[\text{RX}][\text{Nu}^-]$ | Second-order nucleophilic substitution.                                                                                                            |
| <b>Addition pattern</b>                          | $\text{C} = \text{C} + \text{XY} \rightarrow \text{X} - \text{C} - \text{C} - \text{Y}$<br>General addition across a multiple bond.                |
| <b>Elimination pattern</b>                       | $\text{R} - \text{CH}_2 - \text{CH}_2\text{X} \rightarrow \text{R} - \text{CH} = \text{CH}_2 + \text{HX}$<br>Removal of HX to form an alkene.      |
| <b>Aspirin synthesis</b>                         | Salicylic acid + Acetic anhydride $\rightarrow$ Aspirin + Acetic acid<br>Acetylation reaction for aspirin preparation.                             |
| <b>Paracetamol synthesis</b>                     | p-aminophenol + $(\text{CH}_3\text{CO})_2\text{O} \rightarrow$ Paracetamol + $\text{CH}_3\text{COOH}$<br>Selective N-acetylation of p-aminophenol. |

### One-Line Definitions

- **Nucleophile:** Electron-rich species that attacks electron-poor centre.
- **Electrophile:** Electron-deficient species that accepts electron pair.
- **Leaving group:** Group that departs with electron pair.
- **Substitution:** Replacement of one group by another.
- **Addition:** Addition of atoms across a multiple bond.
- **Elimination:** Removal of groups to form a multiple bond.
- **Acetylation:** Introduction of acetyl group,  $\text{CH}_3\text{CO}-$ .

### Frequently Asked / Viva Questions

**Q1. What is an organic reaction mechanism?**

It is the stepwise description of bond breaking and bond formation.

**Q2. Define nucleophile.**

A nucleophile is an electron-rich species that donates an electron pair.

**Q3. Define electrophile.**

An electrophile is an electron-deficient species that accepts an electron pair.

## Frequently Asked / Viva Questions (contd.)

**Q4. Compare  $S_N1$  and  $S_N2$  briefly.**

$S_N1$  is two-step and forms carbocation;  $S_N2$  is one-step backside attack.

**Q5. What is an addition reaction?**

A reaction in which atoms or groups add across a double or triple bond.

**Q6. What is an elimination reaction?**

A reaction in which small molecules are removed to form a multiple bond.

**Q7. How is oxidation recognized in organic chemistry?**

Increase in C-O/C-X bonds or decrease in C-H bonds.

**Q8. How is reduction recognized in organic chemistry?**

Increase in C-H bonds or decrease in C-O/C-X bonds.

**Q9. What is cyclization?**

Intramolecular reaction leading to ring formation.

**Q10. Name reactants used for aspirin synthesis.**

Salicylic acid and acetic anhydride in presence of acid catalyst.

**Q11. Name the starting material and reagents used for Paracetamol synthesis.**

Phenol is the starting material. Reagents are dilute  $HNO_3$  (nitration), followed by steam distillation to isolate the *p*-nitrophenol isomer, then reduction using Fe/HCl or Sn/HCl, and finally acetylation using acetic anhydride.

**Q12. In Paracetamol synthesis, why does acetylation occur at the  $-NH_2$  group rather than the  $-OH$  group of *p*-aminophenol?**

Nitrogen is less electronegative than oxygen, making the lone pair on the amino group ( $-NH_2$ ) more nucleophilic and reactive than the lone pair on the hydroxyl group ( $-OH$ ). This allows selective N-acetylation under controlled conditions.

## Summary Reference Table

| Reaction type       | Core change                 | Exam clue                    |
|---------------------|-----------------------------|------------------------------|
| Substitution        | Group replacement           | Leaving group present        |
| Addition            | Break pi bond, add groups   | Alkene/alkyne reactant       |
| Elimination         | Remove small molecule       | Alkene product               |
| Oxidation/reduction | Change in C-O and C-H bonds | Reagent decides direction    |
| Drug synthesis      | Planned sequence            | Reactants, catalyst, product |